INDEX

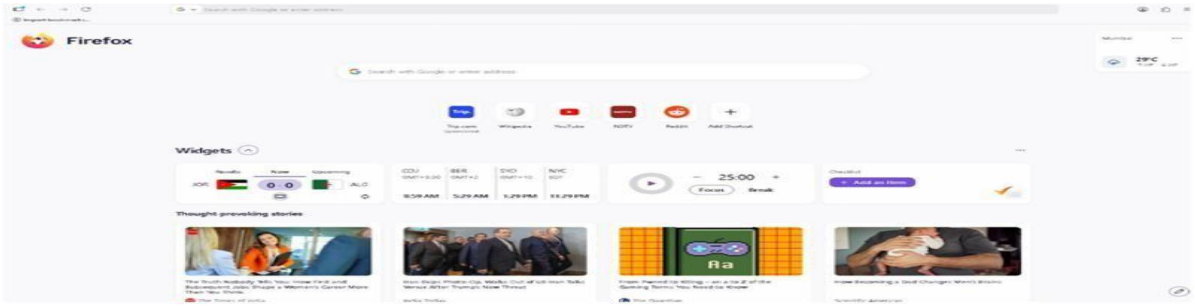
SR. NO	DATE	TITLE	PAGE NO.	SIGN
1		Install Selenium IDE and create a test suite containing a minimum of 4 test cases for different web page formats (e.g., HTML, XML, JSON, etc.).		
2		Conduct a test suite for two different websites using Selenium IDE. Perform various actions like clicking links, filling forms, and verifying content		
3		Install Selenium Server (Selenium RC) and demonstrate its usage by executing a script in Java or PHP to automate browser actions.		
4		Write a program using Selenium WebDriver to automate the login process on a specific web page. Verify successful login with appropriate assertions		
5		Write a program using Selenium WebDriver to update 10 student records in an Excel file. Perform data manipulation and verification.		
6		Write a program using Selenium WebDriver to select the number of students who have scored more than 60 in any one subject (or all subjects). Perform data extraction and analysis.		
7		Write a program using Selenium WebDriver to provide the total number of objects present or available on a web page. Perform object identification and counting		
8		Write a program using Selenium WebDriver to get the number of items in a list or combo box on a web page. Perform element identification and counting		
9		Write a program using Selenium WebDriver to count the number of checkboxes on a web page, including checked and unchecked counts. Perform checkbox identification and counting.		
10		Perform load testing on a web application using JMeter. Generate and analyze load scenarios. Additionally, explore bug tracking using Bugzilla as a tool for logging and tracking software defects.		

PRACTICAL NO. 1

INSTALL SELENIUM IDE; WRITE A TEST SUITE CONTAINING MINIMUM 4 TEST CASES FOR DIFFERENT FORMATS.

Step 1: Install Mozilla Firefox if not Installed

Step 2: Type URL: https://addons.mozilla.org/en-US/firefox/addon/selenium-ide/



=

Step 3: In this SS, where there is a remove button you will have an Add Extension Button instead.

Step 4: Once Mozilla is turned on.

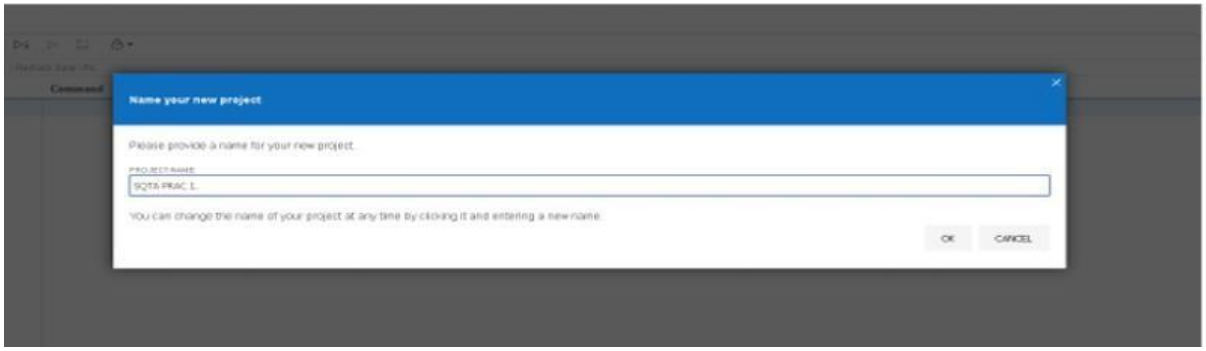
Step 5: Click on the Puzzle Piece like button on the Top-Right Corner as shown in the SS below.

Step 6: A Window like this will pop-up.



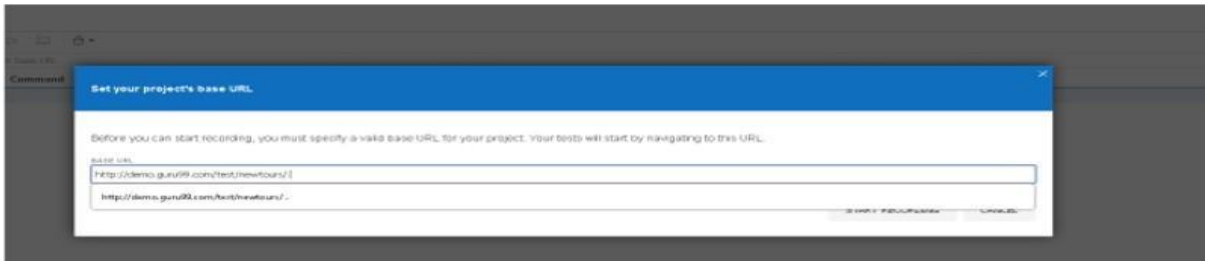
Step 7: Click on “Create a new project”.

Step 8: Give the file name as SQTA Prac1



Step 9: Press “Ok”. Then type the value for our Base URL:
http://demo.guru99.com/test/newtours/

Toggle the Record button on (if it is not yet toggled on by default).



Step 10: A web page like this will open.



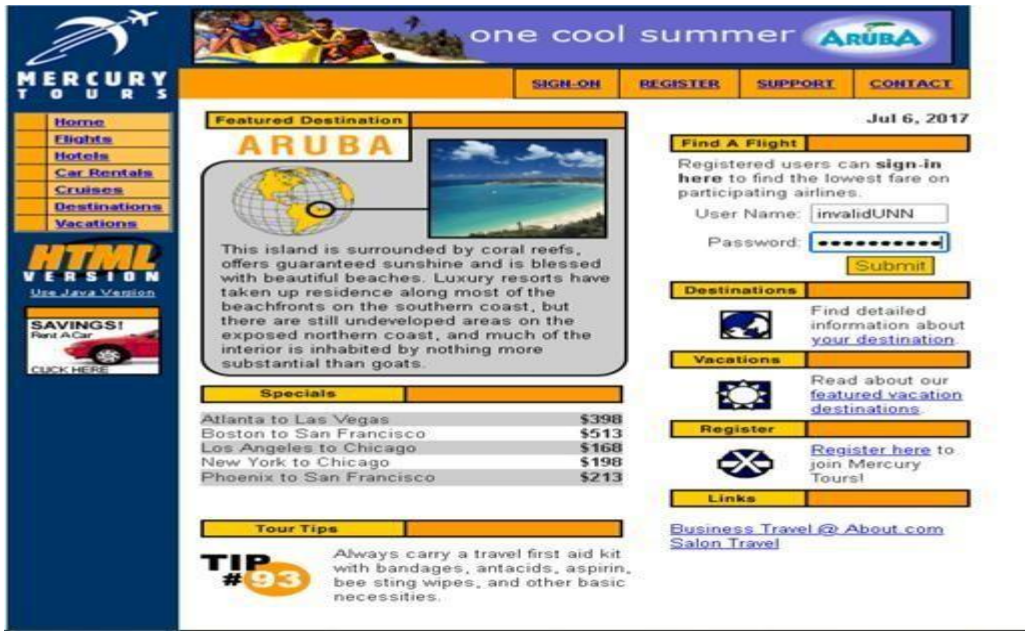
Step 11: Click left on mouse then:

1. Select Selenium IDE
2. Select Assert
3. Select Title



Step 12: In the "User Name" text box of Mercury Tours, type an invalid username, "invalidUNN".

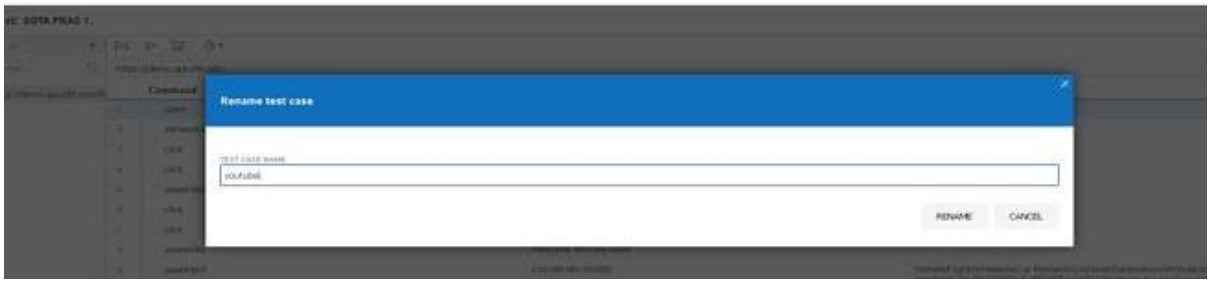
In the "Password" text box, type an invalid password, "invalidPWD".



Step 13: Click on the "Sign-In" button. Firefox should take you to this page.



Step 14: Now that we are done with our test script, we shall save it in a test case.



Step 15: Choose your desired location, and then name the Test Case as "youtube". Click the "Save" button.

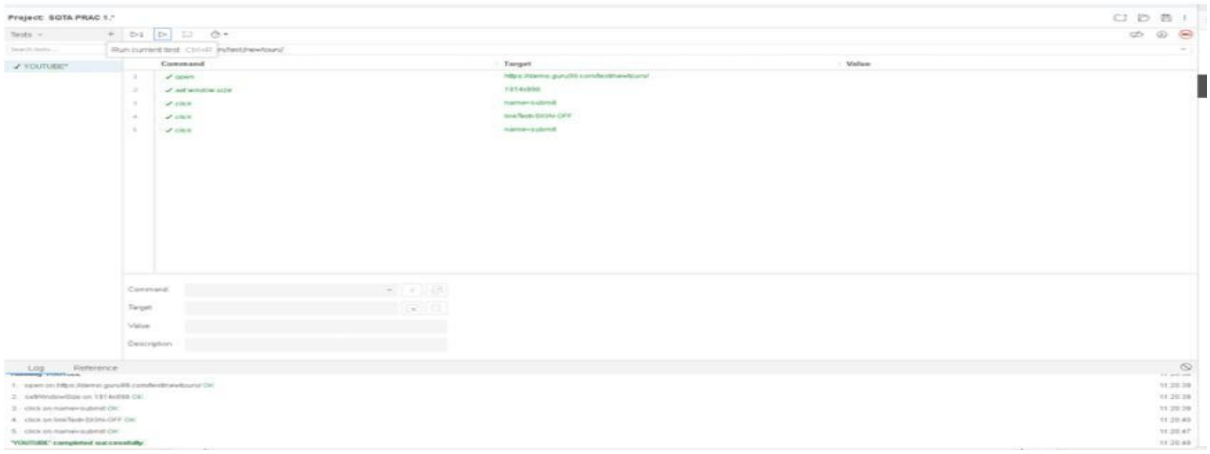
Step 16: Notice that the file was saved as side extension.

Step 17: Go back to Selenium IDE and click the Playback button to execute the whole script. Selenium IDE should be able to replicate everything properly.

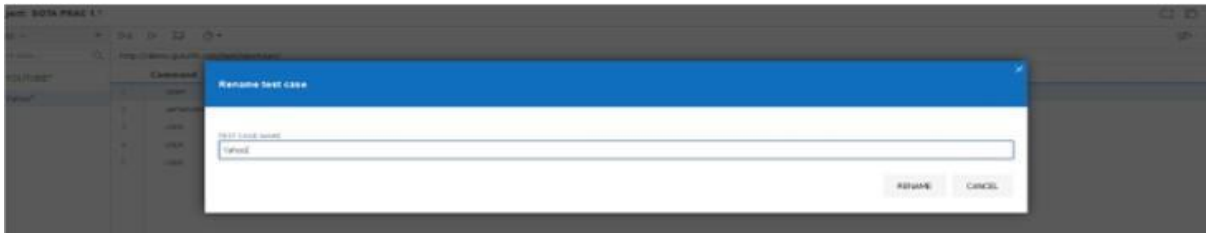
Step 18: Now do this to 3 other websites:

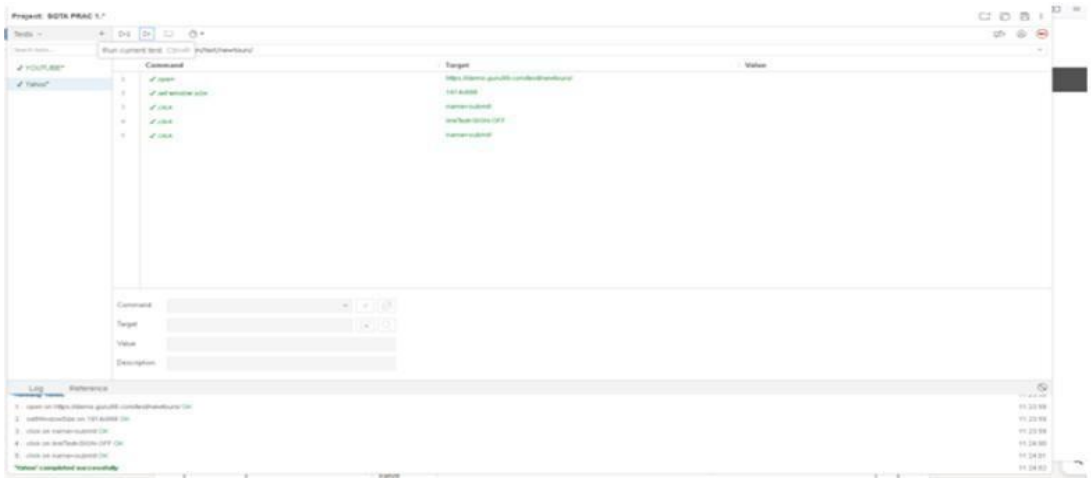
- http://www.youtube.com
- https://www.yahoo.in
- htttp://www.google.com

Step 19: youtube Case

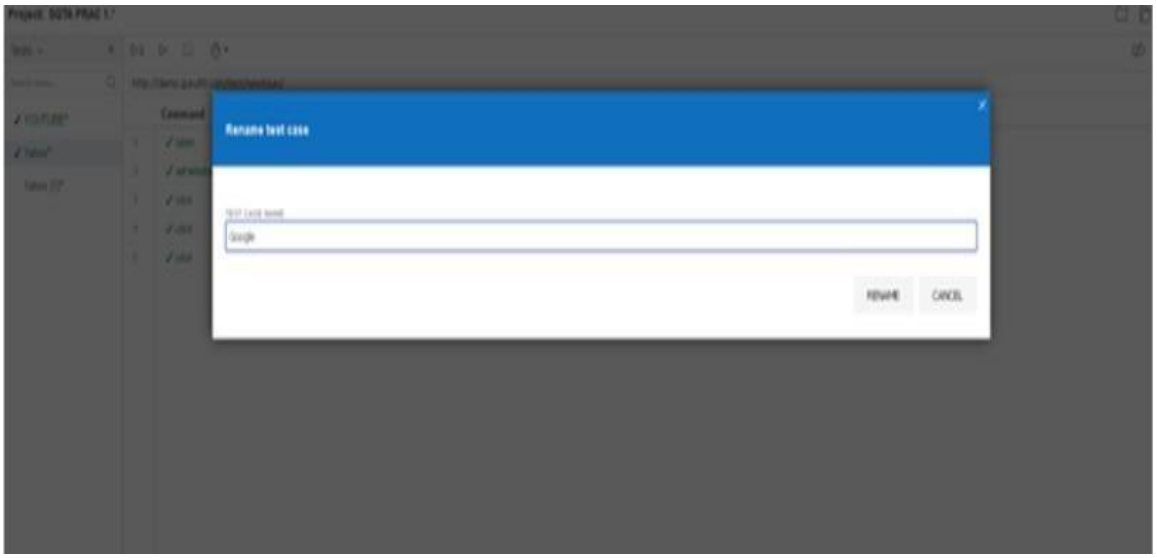


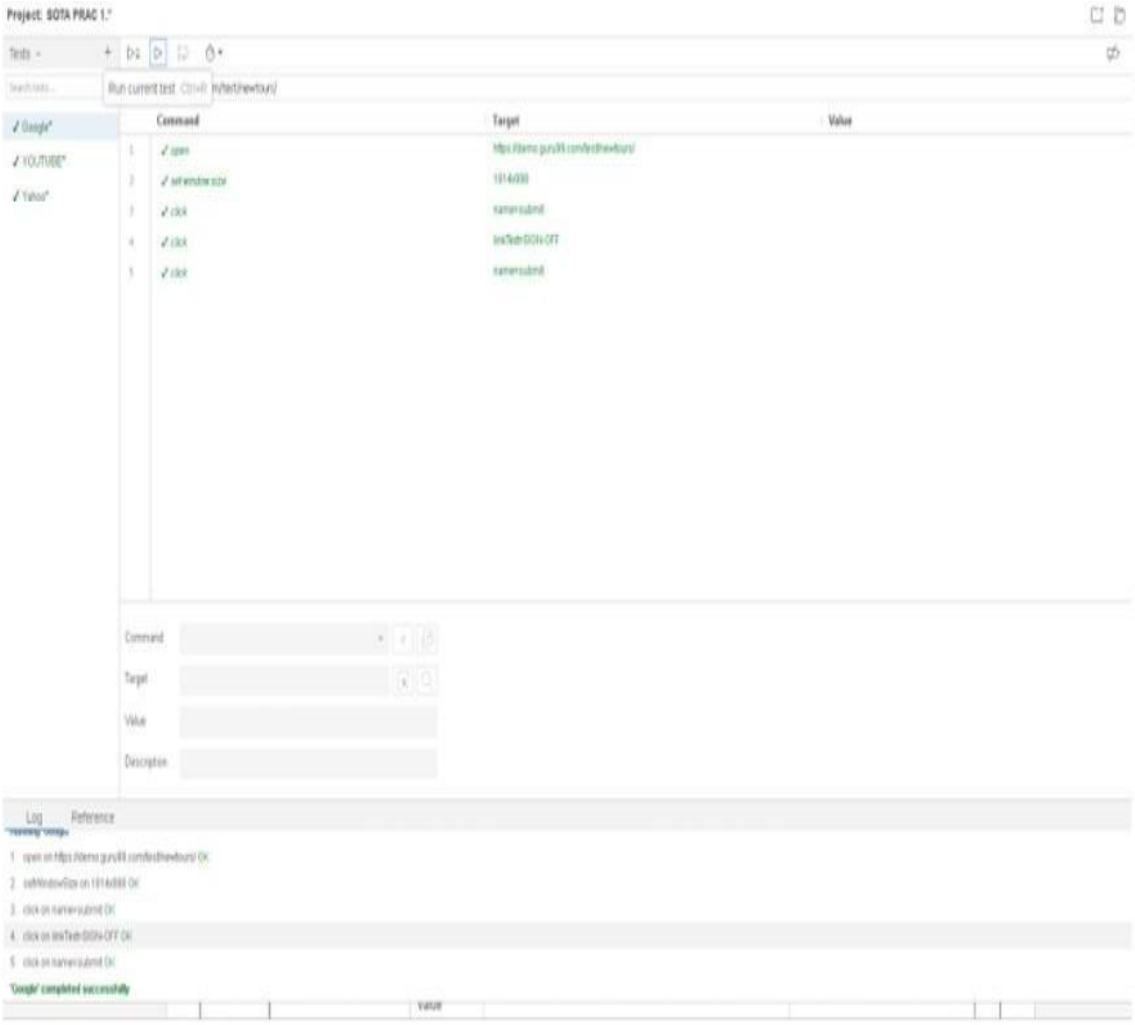
Step 20: Yahoo Case





step 21: Google Case

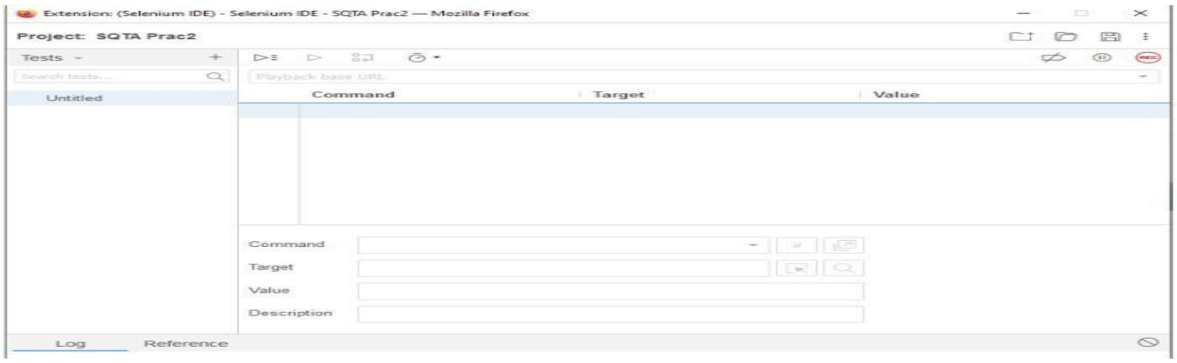

Rename test case
Google
RENAME
CANCEL


Project: SOTA PRAC 1.*
Tests
Run current test Ctrl+R
/webtours/
Google*
YOUTUBE*
Yahoo*
Command
Target
Value
1 open https://demo.guru99.com/webtours/
2 setWindowSize 1014x688
3 click name=submit
4 click linkText=SIGN-OFF
5 click name=submit
Command
Target
Value
Description
Log
Reference
1. open on https://demo.guru99.com/webtours/ OK
2. setWindowSize on 1014x688 OK
3. click on name=submit OK
4. click on linkText=SIGN-OFF OK
5. click on name=submit OK
'Google' completed successfully
Value

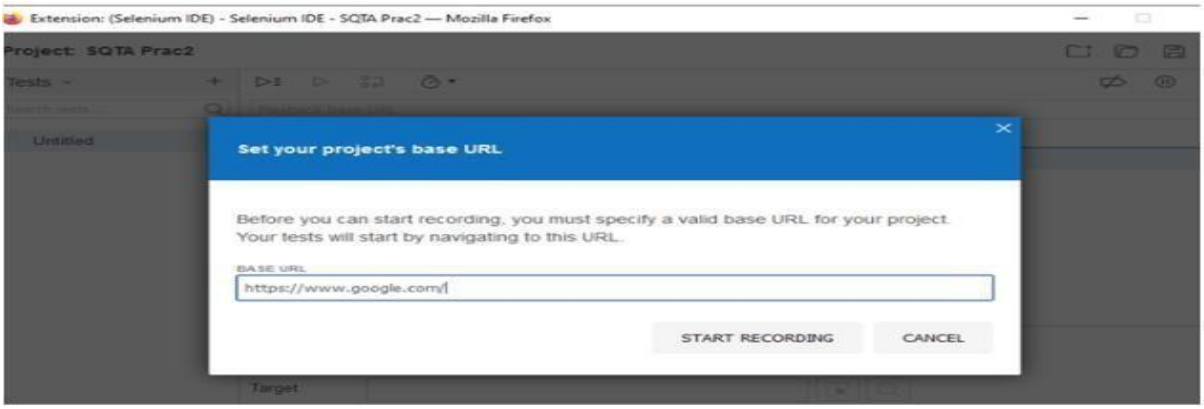
PRACTICAL NO. 2
WRITE 2 TEST CASES.

Step 1: Open Mozilla Firefox.

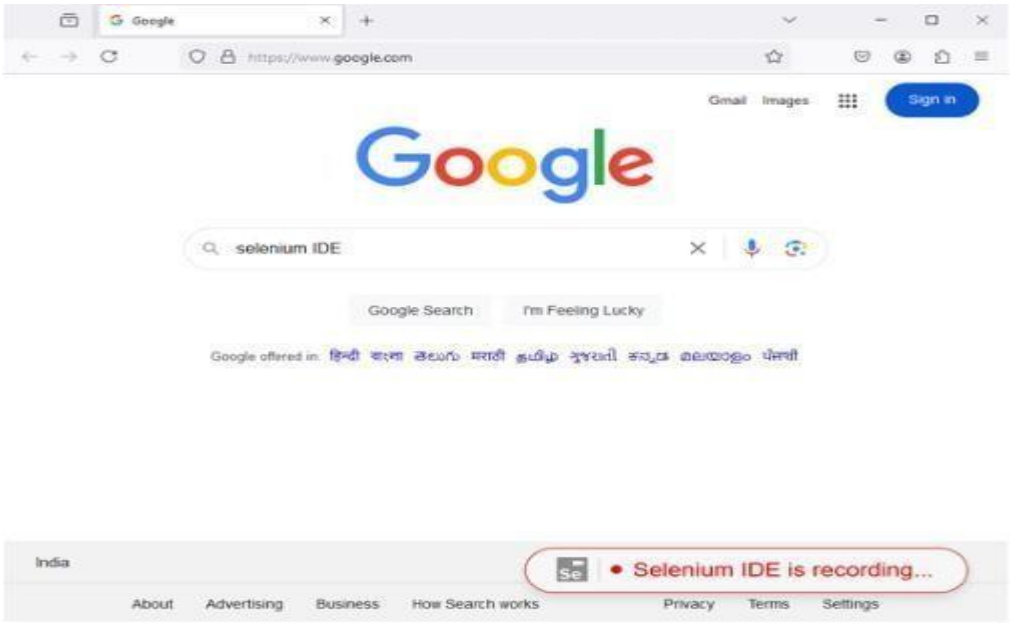
Step 2: Open Selenium IDE.



Step 3: Type this in base URL: https://www.google.com/



Step 4: Search “selenium IDE” in the search bar.

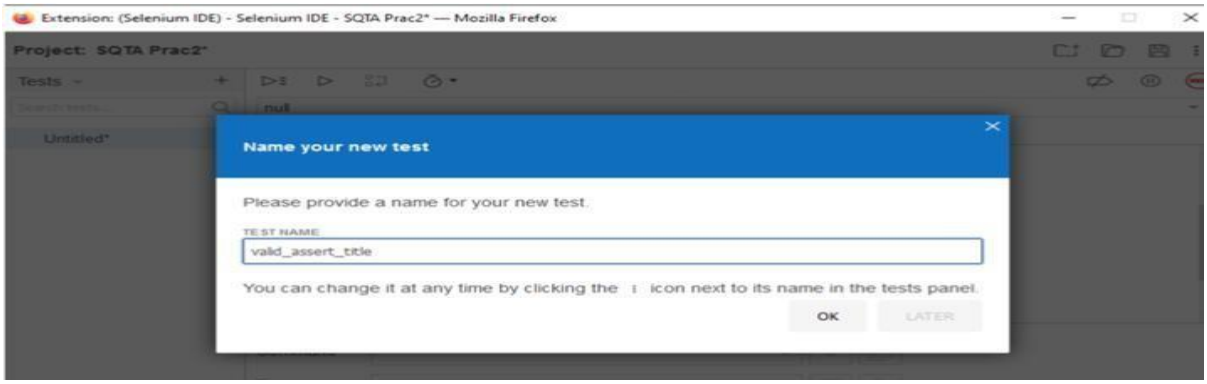


Step 5: Click the first link which takes you to Selenium HQ.

Step 6: Use the Assert Title.

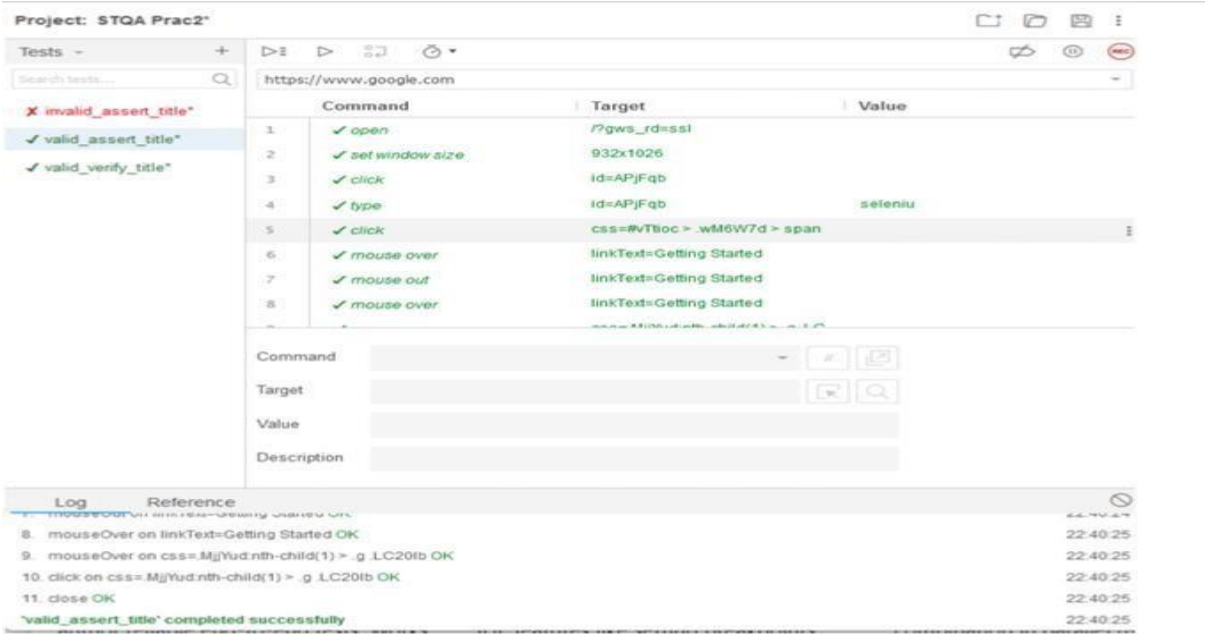


Step 7: Save the test cases's name as "valid_assert_title".



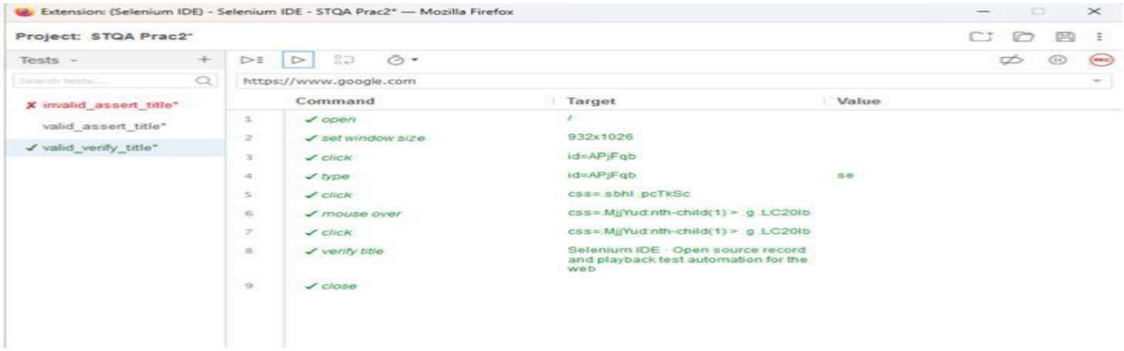
Step 8: Create a new test case named "invalid_assert_title".

Step 9: Now perform from Step 3 to Step 6.



Checking the verify title command:

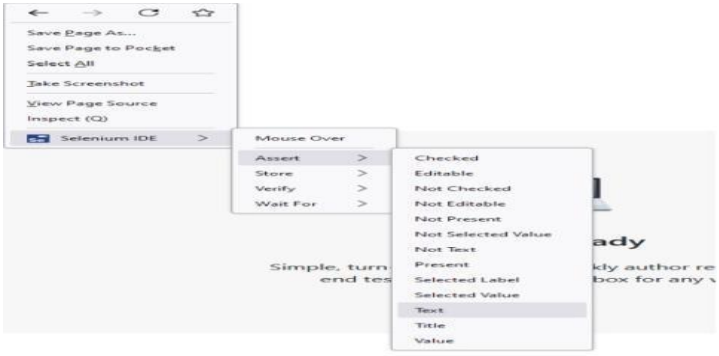
Step 1: Repeat the steps given for assert title command. But, instead of choosing assert title command from Selenium IDE option, choose verify title command.
Give appropriate names to test cases (valid_verify_title and invalid_verify_title).



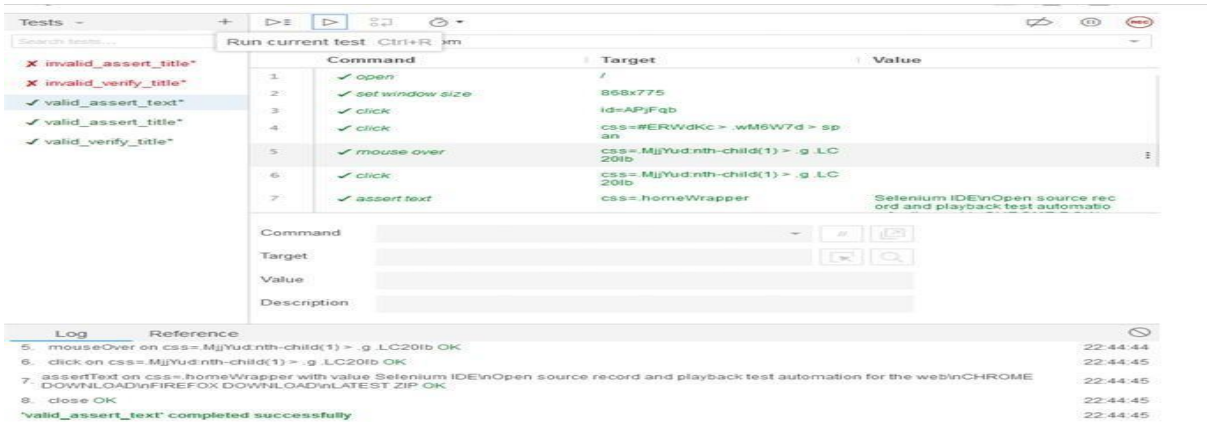
Step 2: Create a new test case named “invalid_verify_title”.
Step 3: Start the recording. Now perform from Step 3 to Step 6. Instead of selecting assert title command, select verify title command.



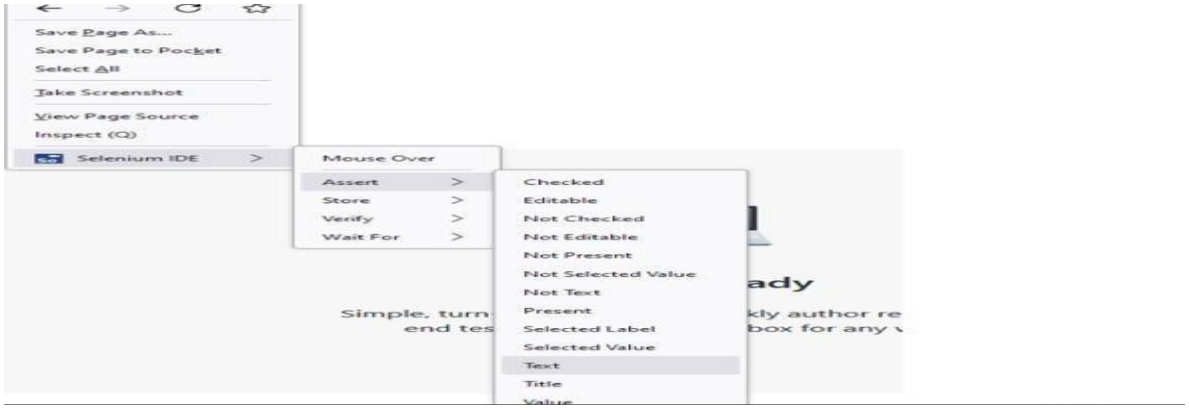
Checking the working of the Assert Text Command:
Step 1: Create a new test named “valid_assert_text”.
Step 2: Start Recording with Base URL same as that of the previous tests.
Step 3: Once the recording starts type in “Selenium IDE”.
Step 4: Click on the first site.
Step 5: Highlight the text shown in the image and then open the Selenium Menu by right clicking mouse button (RMB) and selecting things as shown below.



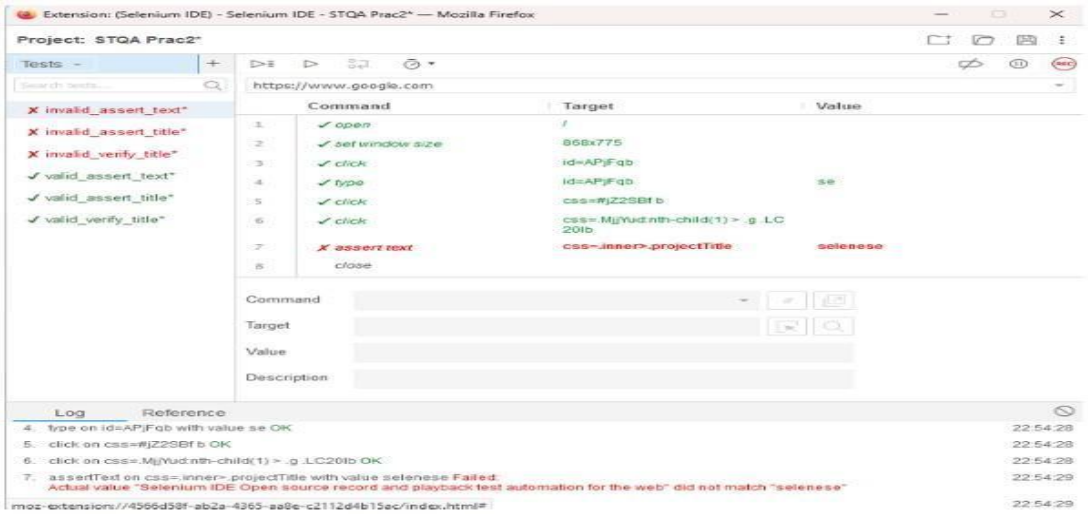
Step 6: Now save the test and stop recording.



Step 7: Open a new test, named “invalid_assert_text”.
Step 8: Do the Steps from Step 1 to Step 4.
Step 9: Highlight the text shown in the image and then open the Selenium Menu by right clicking mouse button (RMB) and selecting things as shown below.

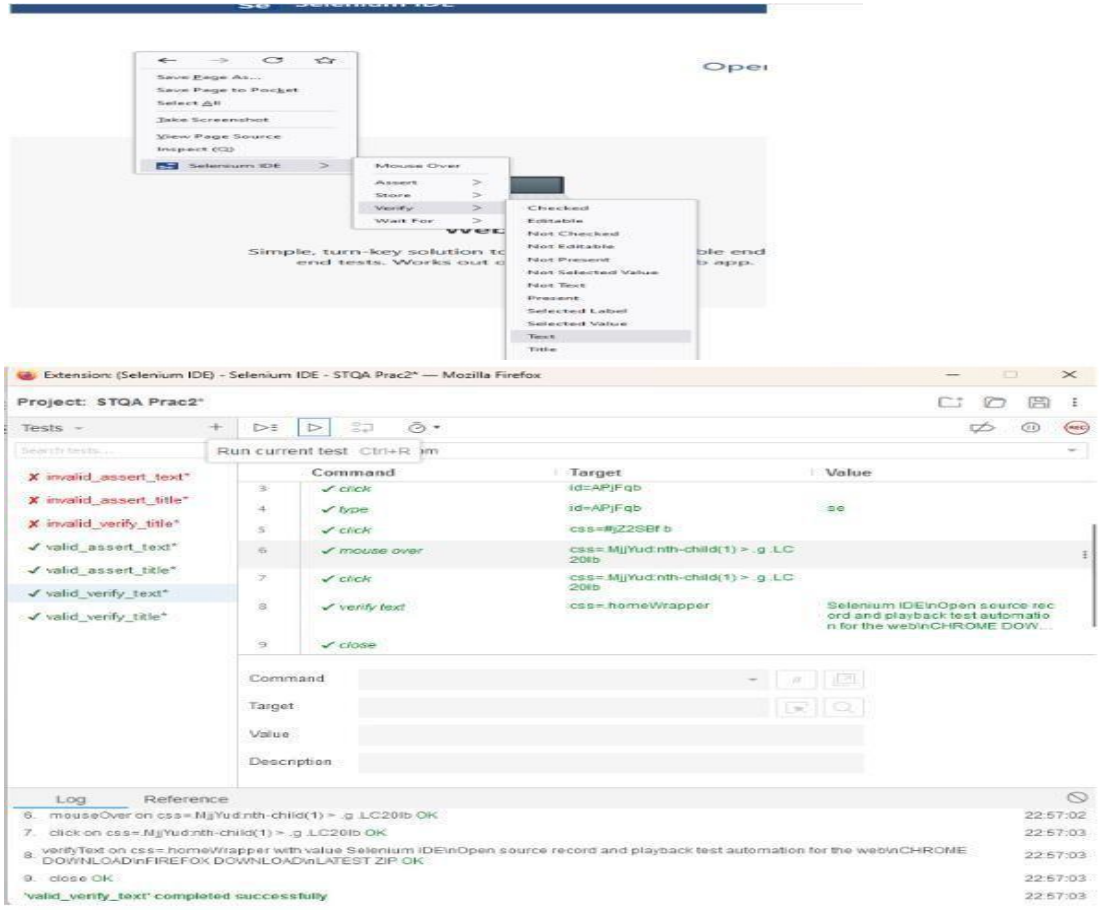


Step 10: Type in the Value Selenese instead of the Selenium IDE in value bar. Step 11: Turn off Recording and save

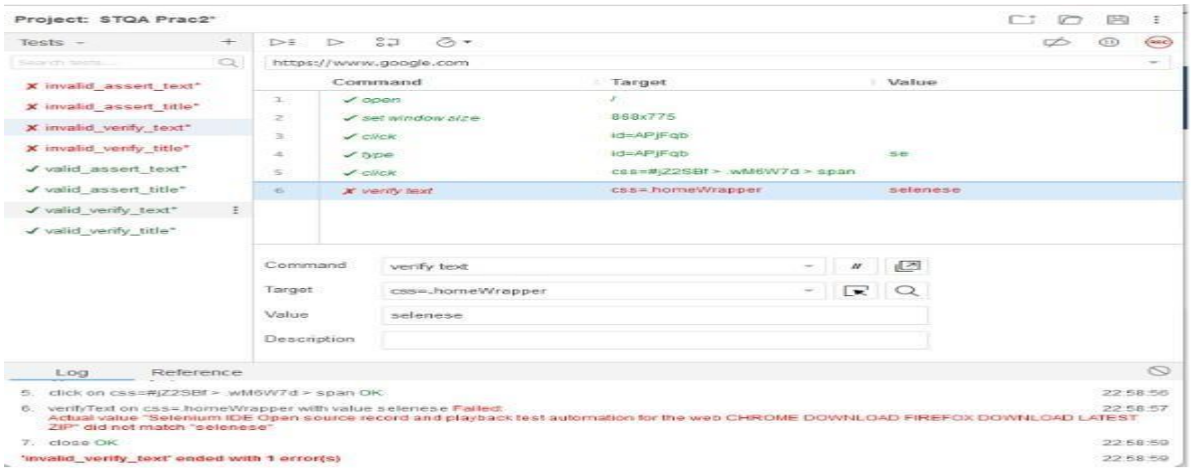


To check the working of verify text command:
Step 1: Repeat the steps given for assert text command. But, instead of choosing assert text command from Selenium IDE option, choose verify text command.

Give appropriate names to test cases.



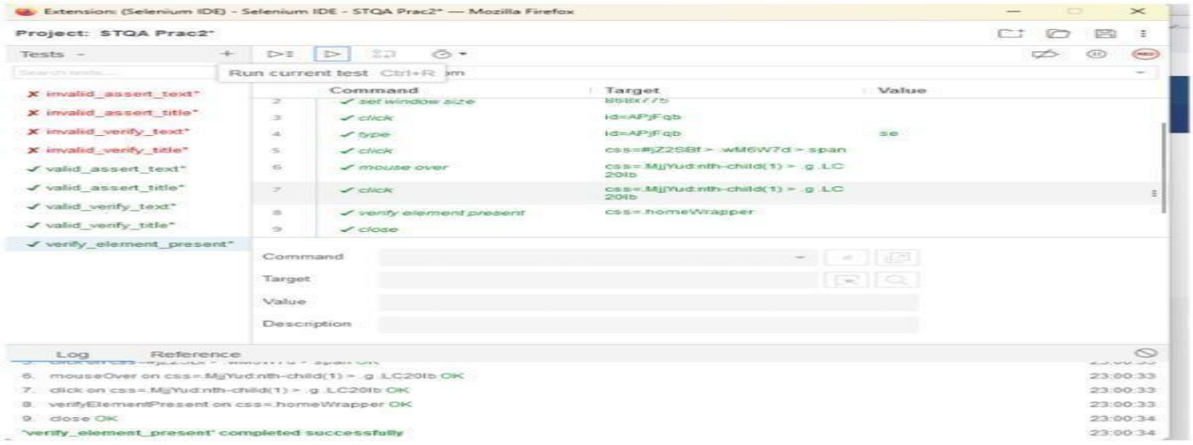
Step 2: Open a new test, named “invalid_verify_text”.
Step 3: Do the Steps from Step 1 to Step 4.
Step 4: Type in the Value Selenese instead of the Selenium IDE in value bar.



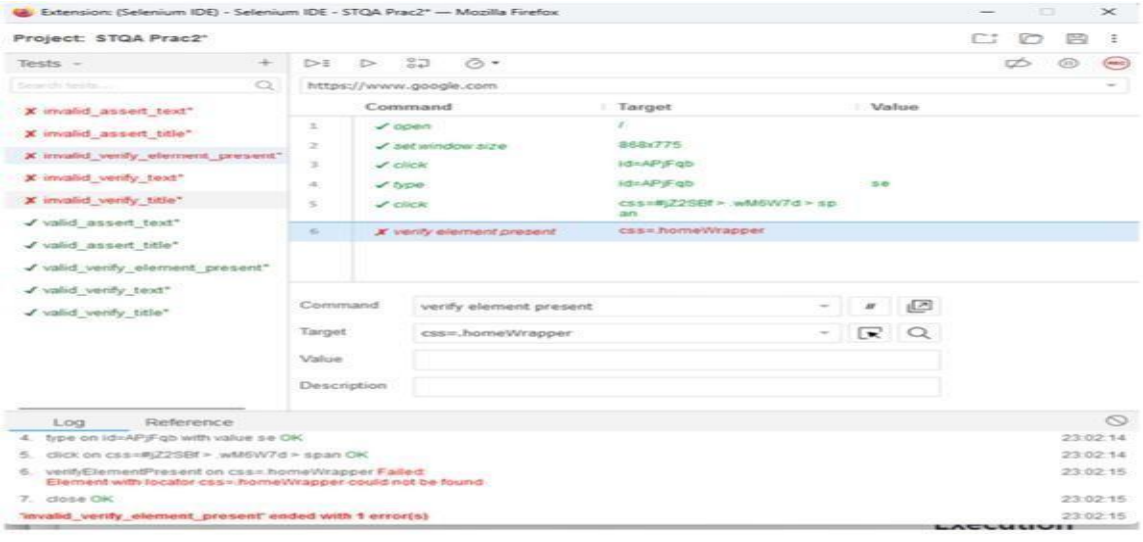
To check the working of verify element present:
Step 1: First create an appropriate test case with an appropriate name like “verify_element_present”.
Step 2: Perform the basic steps.



Step 4: Stop Recording and save then Run the Test.



Step 5: Now create a new test named “invalid_verify_element_present”. Step 6: Do as done in the SS Below.



PRACTICAL NO. 3

INSTALL SELENIUM SERVER (SELENIUM RC) AND DEMONSTRATE ITS USAGE BY EXECUTING A SCRIPT IN JAVA OR PHP TO AUTOMATE BROWSER ACTIONS.

Step 1: Install Java. Download Java SE Development Kit 16.0.2 according to the Windows platform being used.

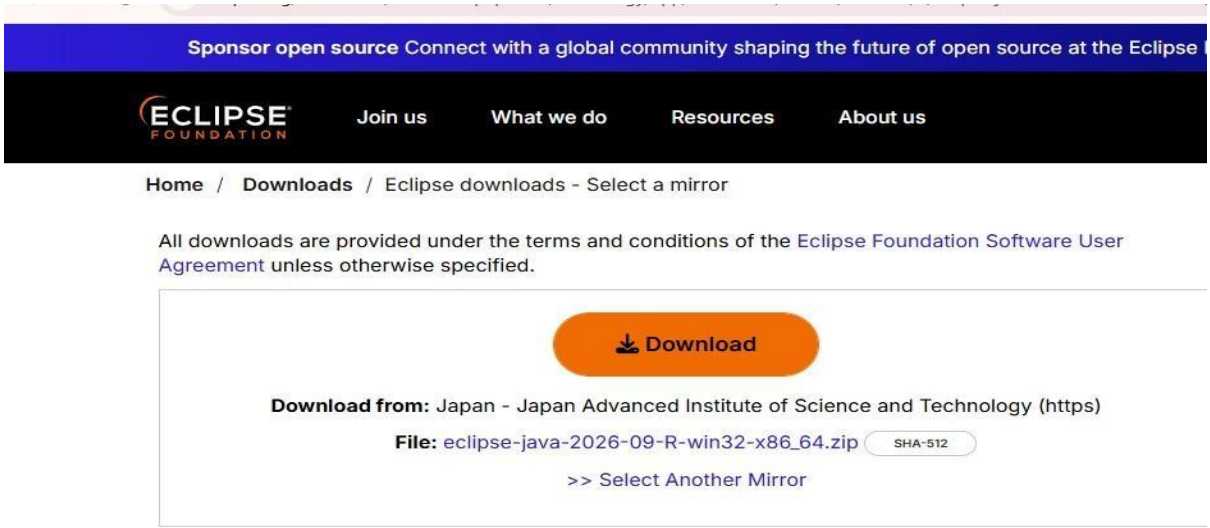

JDK 27 will receive updates under these terms, until March 2027, when it will be superseded by JDK 28.

Linux macOS Windows

Product/file description	File size	Download
x64 Compressed Archive	209.82 MB	https://download.oracle.com/java/27/latest/jdk-27_windows-x64_bin.zip (sha256)
x64 Installer	187.54 MB	https://download.oracle.com/java/27/latest/jdk-27_windows-x64_bin.exe (sha256)
x64 MSI Installer	186.32 MB	https://download.oracle.com/java/27/latest/jdk-27_windows-x64_bin.msi (sha256)

Step 2: Download Windows x64 Installer. Run the JDK Installer by double-clicking on the file name in the download location and following the instructions on the instruction wizard.

Step 3: Install Eclipse IDE.



Step 4: Download and Install Selenium to be set up in Eclipse.

Step 5: Add JDK zip file to its path.

Chrome Driver (for Chrome),

Gecko Driver (for Firefox),

Safari Driver (for Safari),

Internet Explorer Driver,

MS Edge Driver (IE and Edge respectively).

Place these Browser Driver files in a directory that is part of the environment PATH.

Step 6: Install Java Language Bindings.

• Version 3.141.59 (2018)

• Changelog

• API Docs

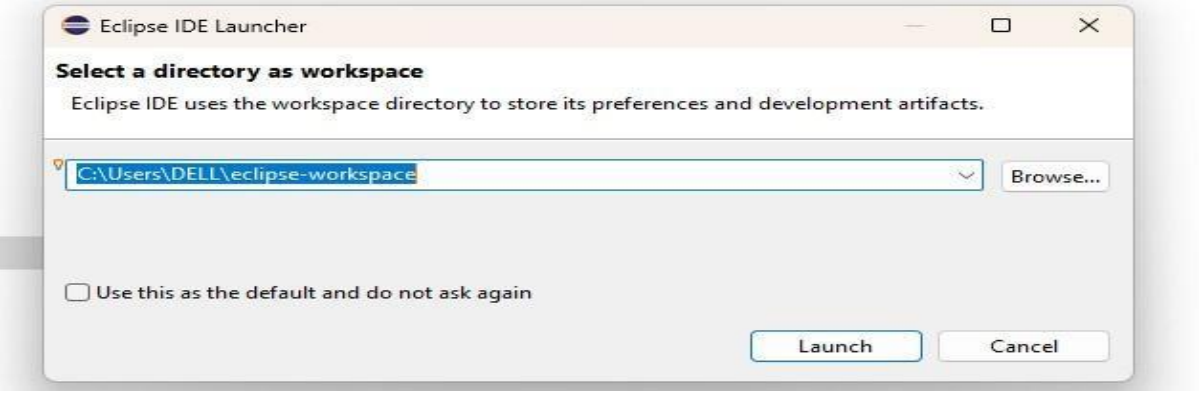


Step 7: Install Eclipse.

Step 8: Launch Eclipse. To launch Eclipse double click on the eclipse.exe file in the download location.

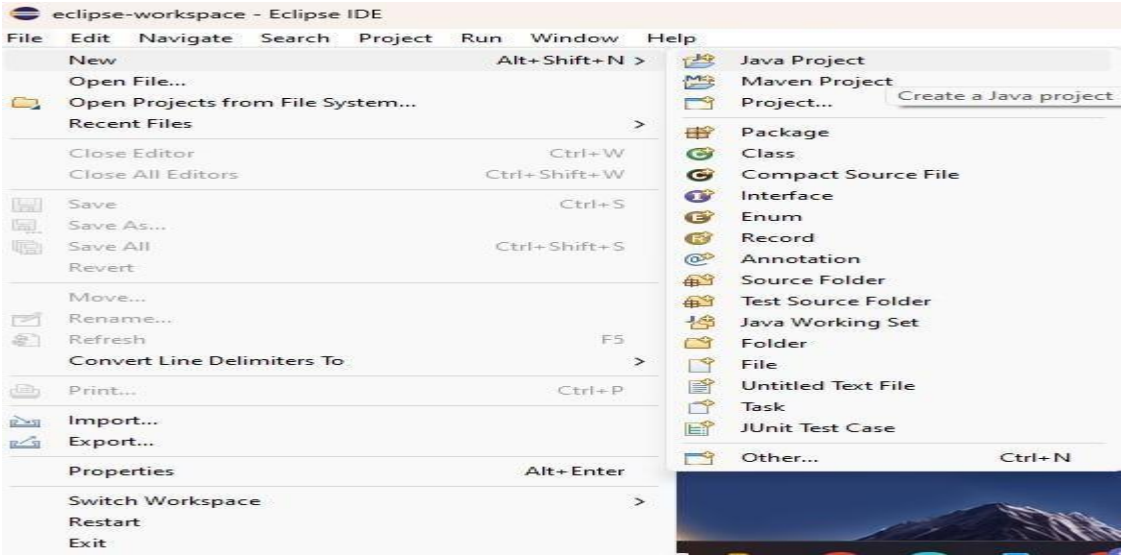
Step 9: Create Workspace in Eclipse.

This workspace named “C:\BrowserStack” is like any other folder, which will store all the test scripts. Launch the BrowserStack workspace.



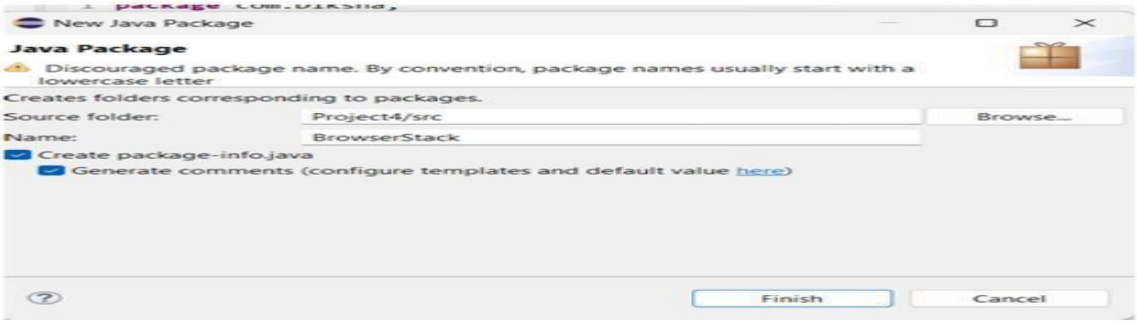
Step 10: Create New Java Project in the Browser-Stack Workspace.

Create a new Java Project by clicking on File > New > Java Project and name it.

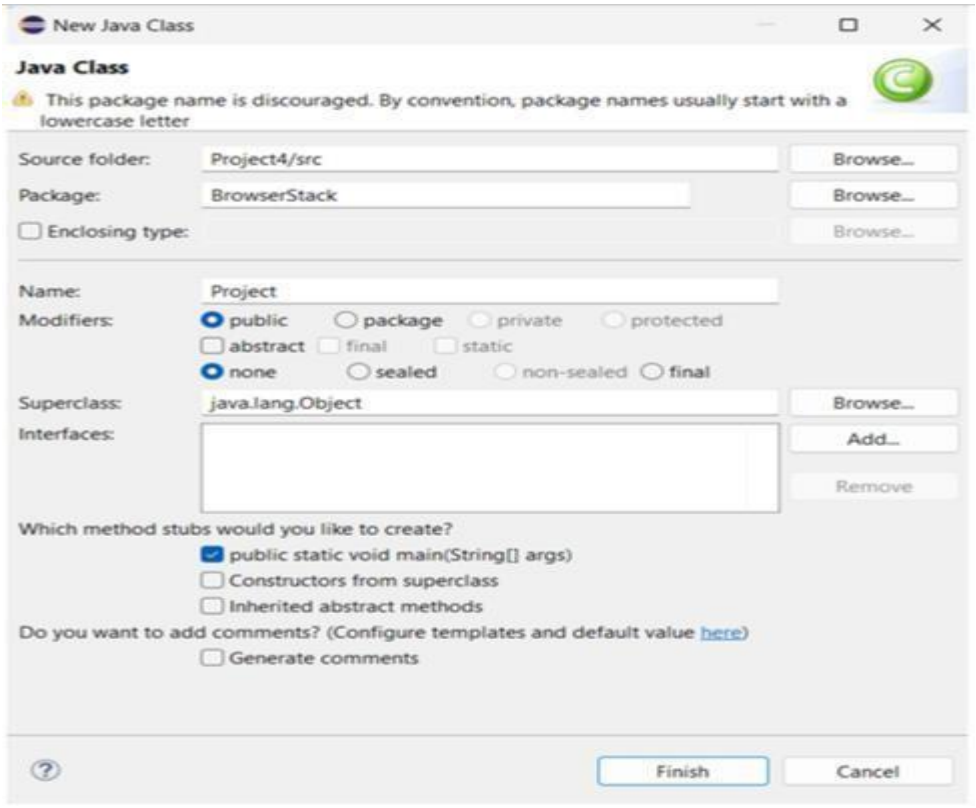


Step 11: Create Package and Class under the Java Project.

By clicking on the Src folder (which is the source folder), create a new package and name it (Browser-Stack). Then right-click on the package name and create a class.



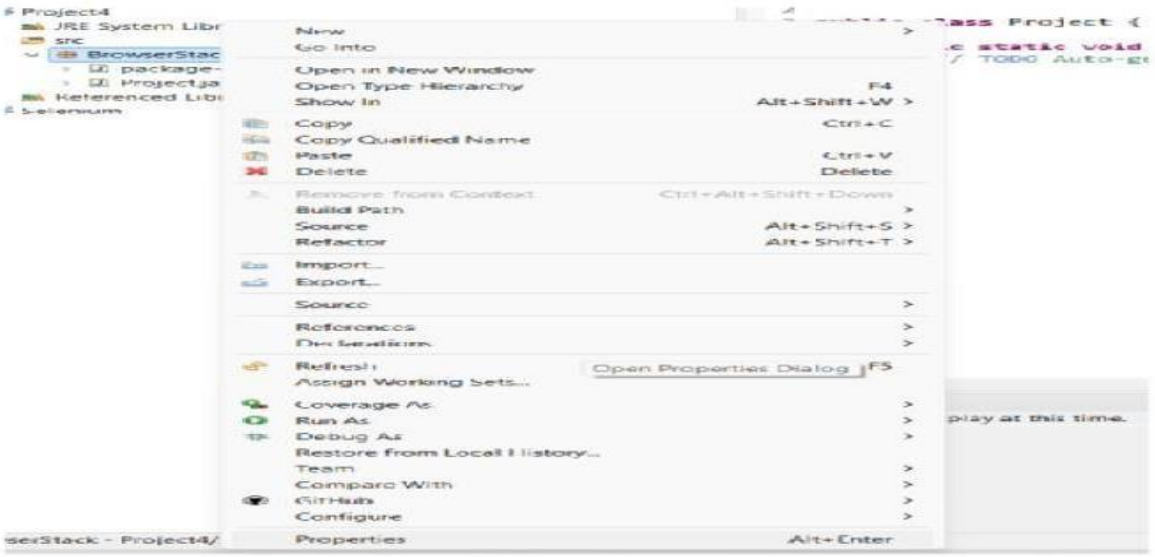
Step 12: Creating Class in the BrowserStack Package. And select public static void option.

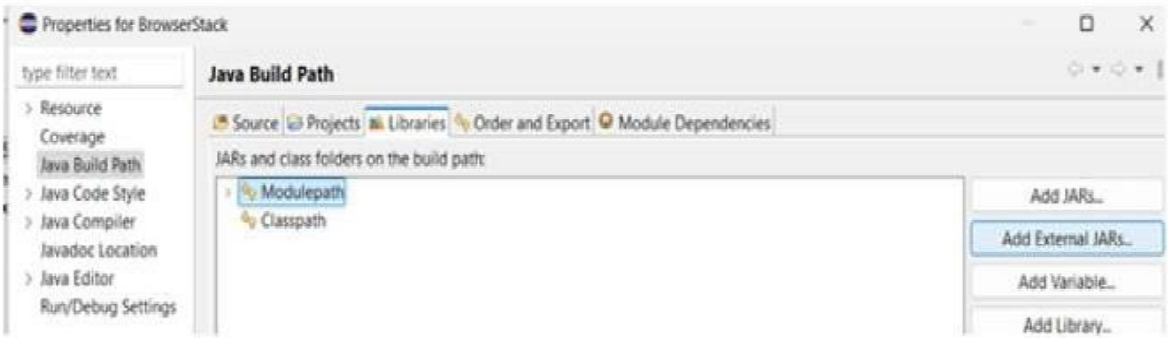


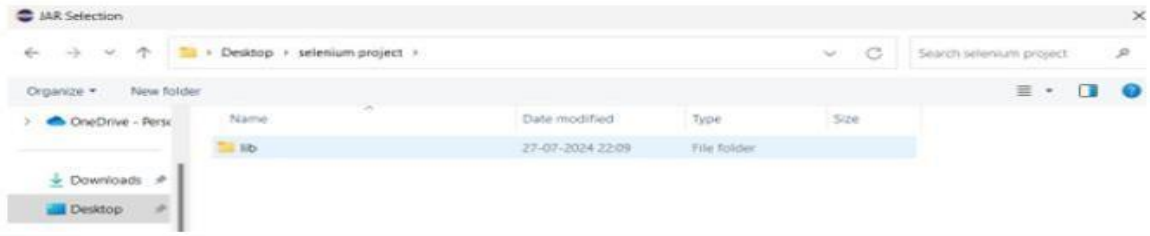
Step 13: Add Selenium JARs to the Java Project in Eclipse.

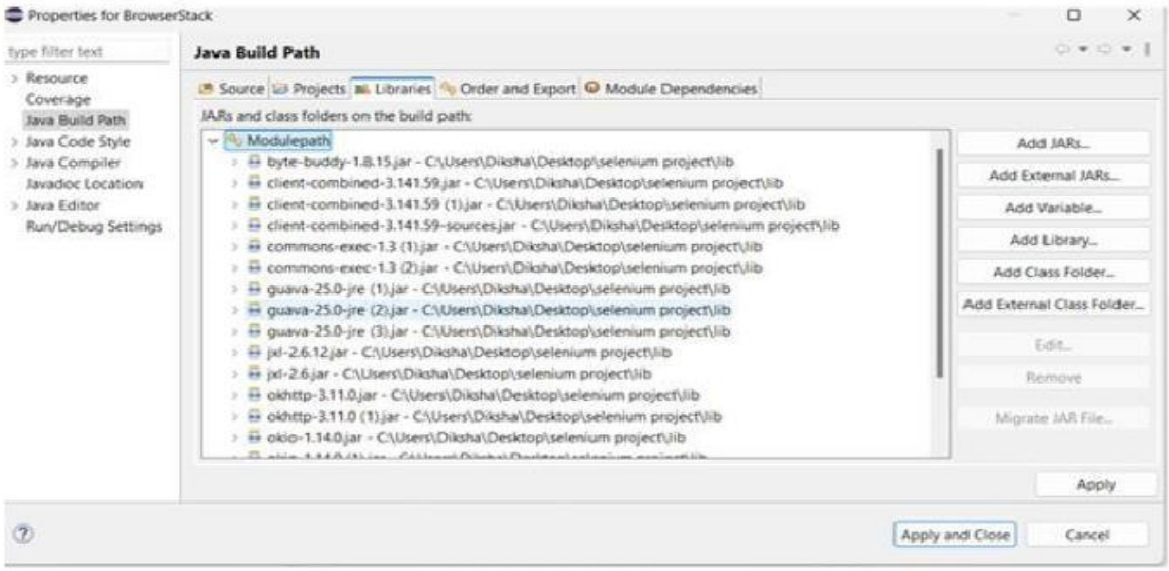
To add the Selenium Jars to the Browser-Stack Java right click on the Browser-Stack Project folder and select the Properties option.

In the properties window, click on the Java Build Path and Add External JARs. Browse and add the downloaded Selenium JARs i.e. Client Combined JAR and all the JARs under the Libs folder, then click Apply and Close.


Project4
JRE System Libr
src
BrowserStac
package-
Project.ja
Referenced Libr
Selenium
New
Go Into
Open in New Window
Open Type Hierarchy F4
Show In Alt+Shift+W
Copy Ctrl+C
Copy Qualified Name
Paste Ctrl+V
Delete Delete
Remove from Context Ctrl+Alt+Shift+Down
Build Path
Source Alt+Shift+S
Refactor Alt+Shift+T
Import...
Export...
Source
References
Refresh F5
Open Properties Dialog
Assign Working Sets...
Coverage As
Run As
Debug As
Restore from Local History...
Team
Compare With
GitHub
Configure
Properties Alt+Enter


Properties for BrowserStack
type filter text
Resource
Coverage
Java Build Path
Java Code Style
Java Compiler
Javadoc Location
Java Editor
Run/Debug Settings
Java Build Path
Source
Projects
Libraries
Order and Export
Module Dependencies
JARs and class folders on the build path:
Modulepath
Classpath
Add JARs...
Add External JARs...
Add Variable...
Add Library...


JAR Selection
Desktop
selenium project
Search selenium project
Organize
New folder
OneDrive - Pers
Downloads
Desktop
Name
Date modified
Type
Size
lib
27-07-2024 22:09
File folder


Properties for BrowserStack
type filter text
Resource
Coverage
Java Build Path
Java Code Style
Java Compiler
Javadoc Location
Java Editor
Run/Debug Settings
Java Build Path
Source
Projects
Libraries
Order and Export
Module Dependencies
JARs and class folders on the build path:
Modulepath
byte-buddy-1.8.15.jar - C:\Users\Diksha\Desktop\selenium project\lib
client-combined-3.141.59.jar - C:\Users\Diksha\Desktop\selenium project\lib
client-combined-3.141.59 (1).jar - C:\Users\Diksha\Desktop\selenium project\lib
client-combined-3.141.59-sources.jar - C:\Users\Diksha\Desktop\selenium project\lib
commons-exec-1.3 (1).jar - C:\Users\Diksha\Desktop\selenium project\lib
commons-exec-1.3 (2).jar - C:\Users\Diksha\Desktop\selenium project\lib
guava-25.0-jre (1).jar - C:\Users\Diksha\Desktop\selenium project\lib
guava-25.0-jre (2).jar - C:\Users\Diksha\Desktop\selenium project\lib
guava-25.0-jre (3).jar - C:\Users\Diksha\Desktop\selenium project\lib
jxl-2.6.12.jar - C:\Users\Diksha\Desktop\selenium project\lib
jxl-2.6.jar - C:\Users\Diksha\Desktop\selenium project\lib
okhttp-3.11.0.jar - C:\Users\Diksha\Desktop\selenium project\lib
okhttp-3.11.0 (1).jar - C:\Users\Diksha\Desktop\selenium project\lib
okio-1.14.0.jar - C:\Users\Diksha\Desktop\selenium project\lib
Add JARs...
Add External JARs...
Add Variable...
Add Library...
Add Class Folder...
Add External Class Folder...
Edit...
Remove
Migrate JAR File...
Apply
Apply and Close
Cancel

PRACTICAL NO. 4

WRITE AND TEST A PROGRAM TO LOGIN SPECIFIC WEB PAGE. CREATE A NEW JAVA PROJECT AND ALSO ADD THE RESPECTIVE .JAR FILES AND ALSO ADD A NEW CLASS. THEN TYPE THE FOLLOWING CODE:

```
package Prac3;  import org.openqa.selenium.By; import

org.openqa.selenium.WebDriver; import

org.openqa.selenium.firefox.FirefoxDriver;  public class P3 {

public static void main(String[] args) throws InterruptedException {

      System.setProperty(

         "webdriver.gecko.driver",

         "D:\\STQA\\Mozilla Driver-20230927T030950Z-001\\Mozilla
Driver\\geckodriver.exe"

      );

      WebDriver driver = new FirefoxDriver();

Driver.get("https://accounts.google.com/");          driver.findElement(By.id("identifierId"))

          .sendKeys("debo123@gmail.com"); // enter username

driver.findElement(By.id("identifierNext")).click();

Thread.sleep(500)        driver.findElement(By.name("password"))

           .sendKeys("TYPE_YOUR_PASSWORD"); // enter password

driver.findElement(By.id("passwordNext")).click();         String title =

driver.getTitle()        if (title.equals("Google Account")) {

         System.out.println("LOGIN SUCCESSFUL...");
      } else {

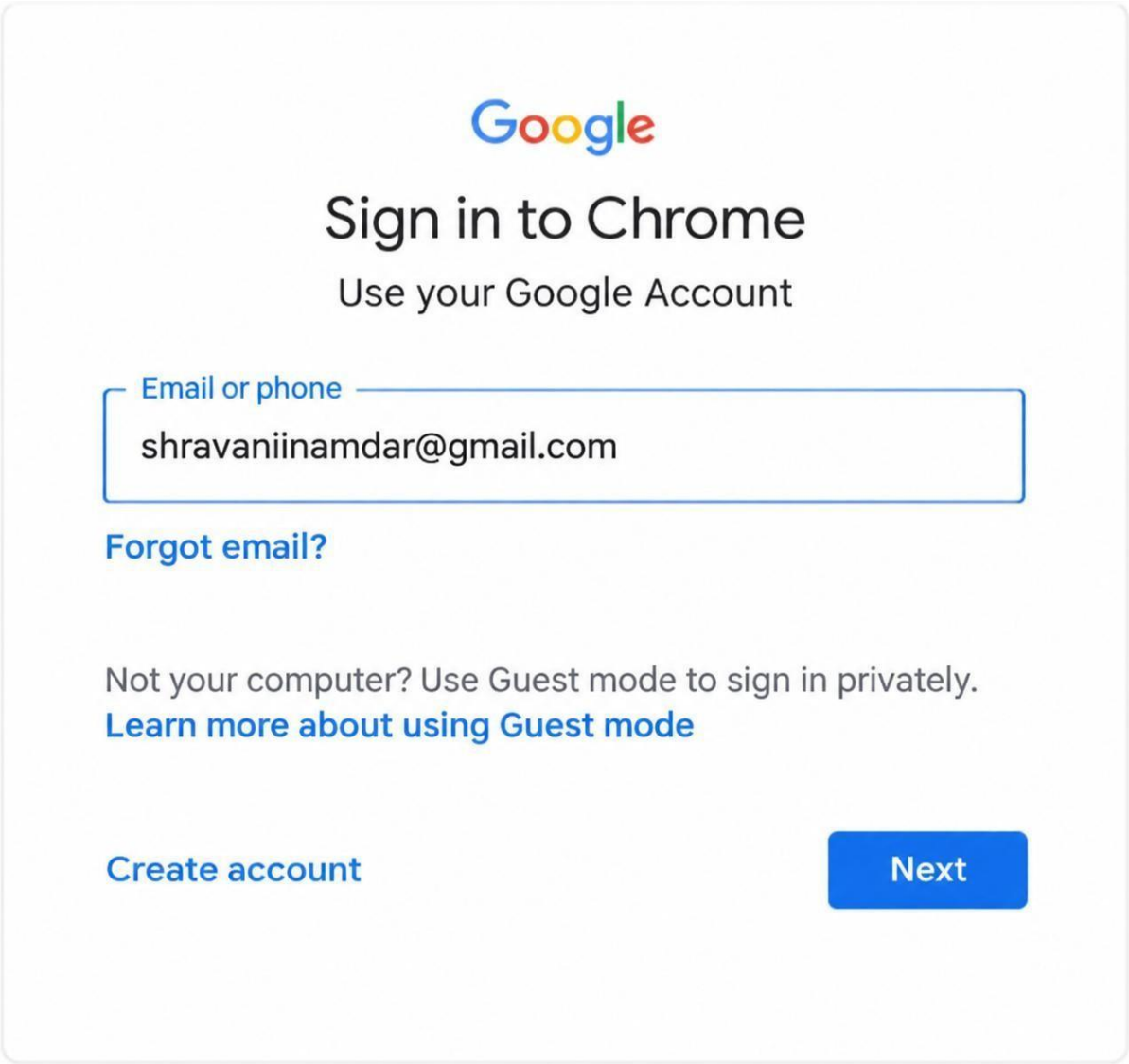
         System.out.println("LOGIN FAILED");

      }

   }

}
```

OUTPUT:


Google
Sign in to Chrome
Use your Google Account
Email or phone
shravaniinamdar@gmail.com
Forgot email?
Not your computer? Use Guest mode to sign in privately.
Learn more about using Guest mode
Create account
Next

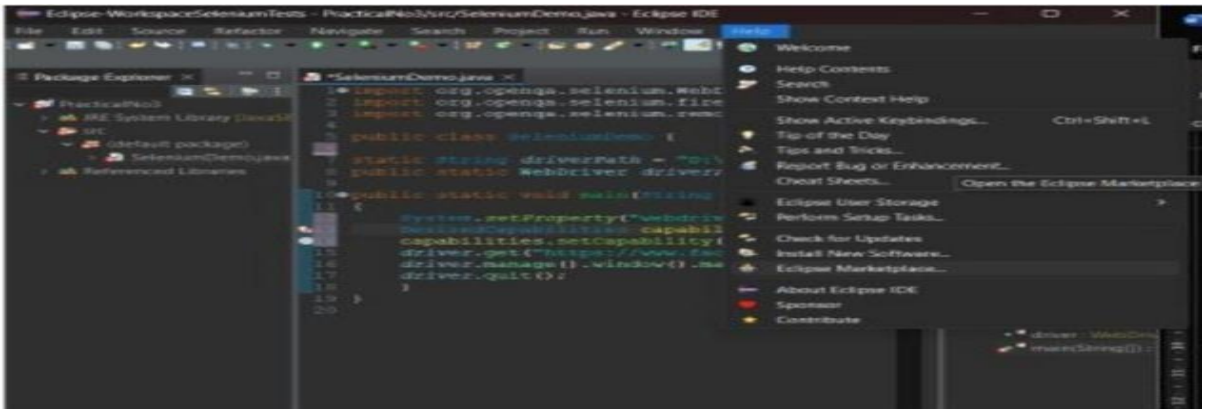
PRACTICAL NO. 5
WRITE AND TEST A PROGRAM TO UPDATE 10 STUDENT RECORDS INTO TABLE INTO EXCEL FILE.

Step 1: Open The Google And Download The Jxl Jar file. Use This URL: https://sourceforge.net/projects/jexc

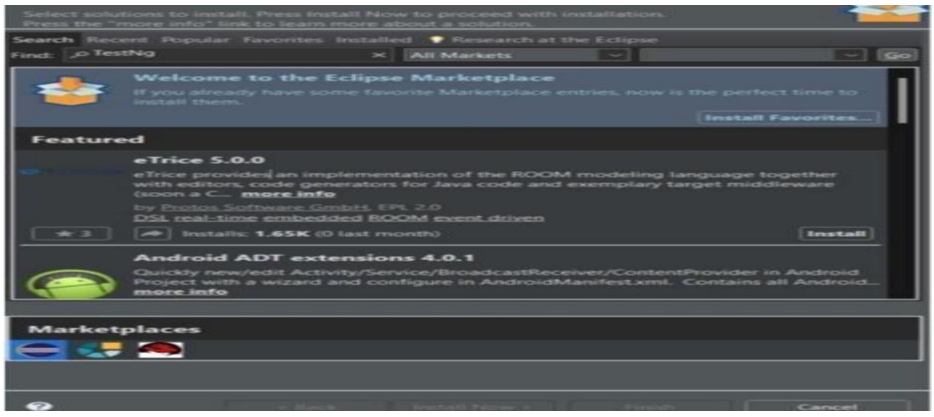
Step 2: We Need to Create the Excel file from the Students records present in it. Eclipse Does not support .xlsx files; save your file in .xls extension only.

Roll No	Java	DBMS	OS	Total
1	34	45	23	102
2	64	34	34	135
3	56	45	56	157
4	89	76	76	241
5	65	98	56	219
6	45	77	67	189
7	35	55	43	133
8	67	57	34	158
9	54	76	56	186
10	78	56	33	167

Step 3: Now We need to install the TestNG Plugins For The Eclpise IDE. 1. Open the Eclipse IDE , Go to the Help option and click on Eclipse Marketplace.



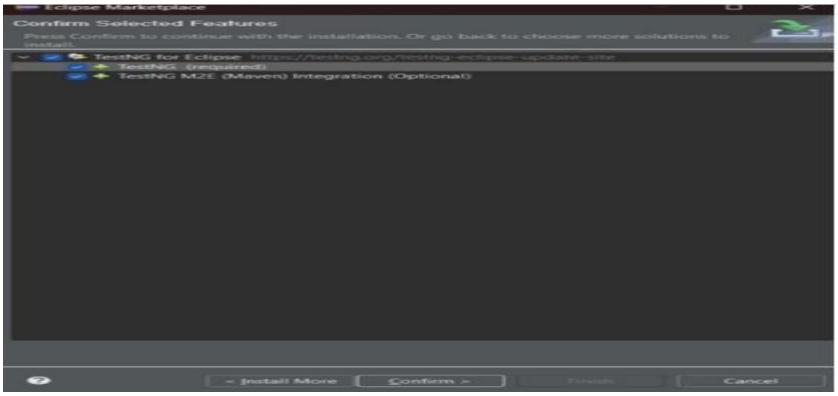
2. Click on the search bar and search the TestNG.



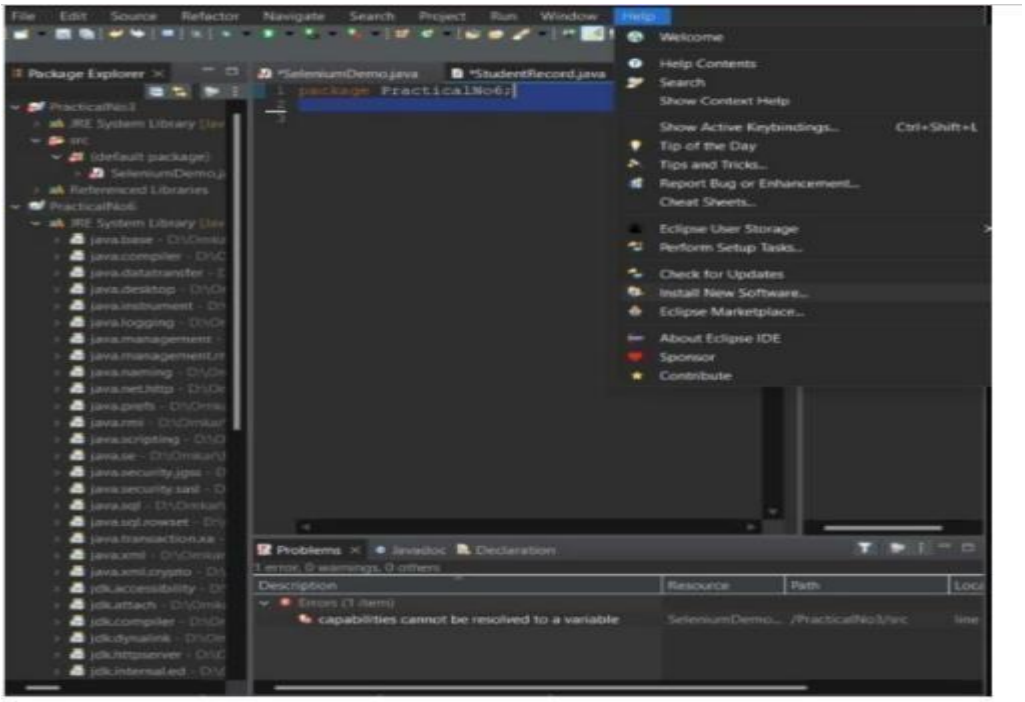
3. Then Install the TestNG For Eclipse.



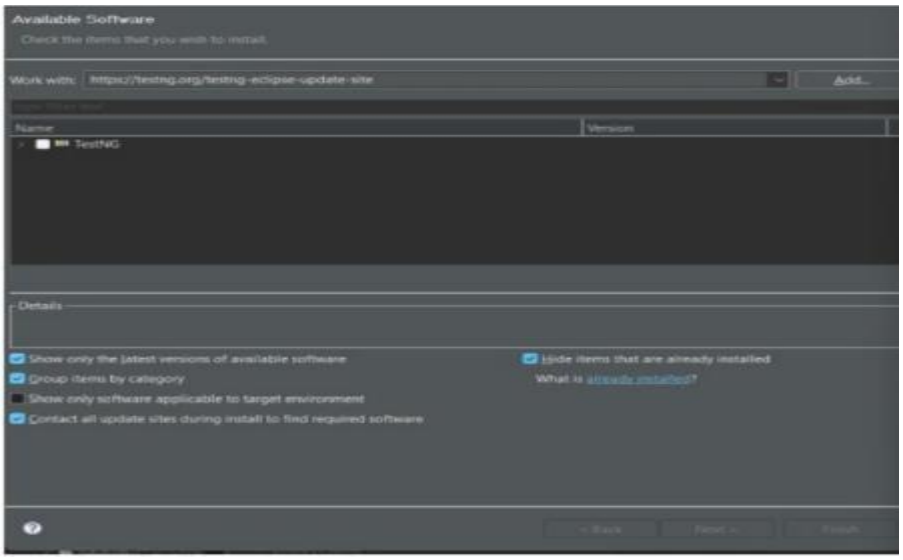
4. Now then click the Confirm And It's Finish the installation.



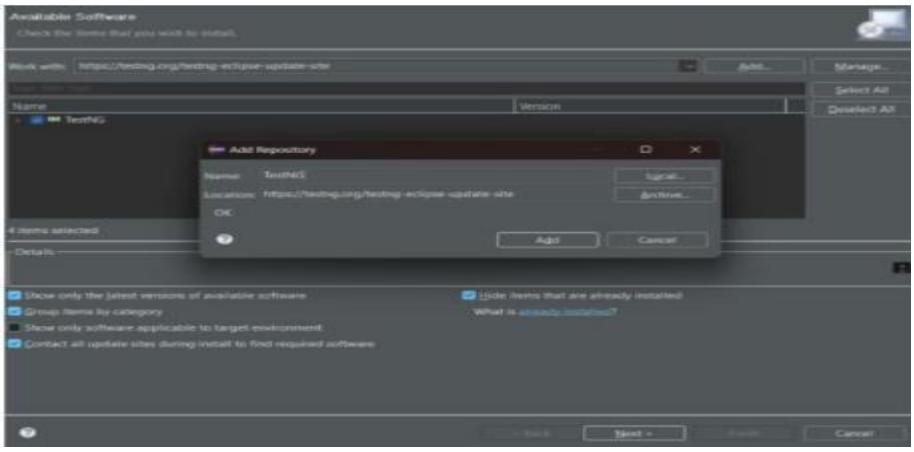
5. Go to help Option Select the install new software option.



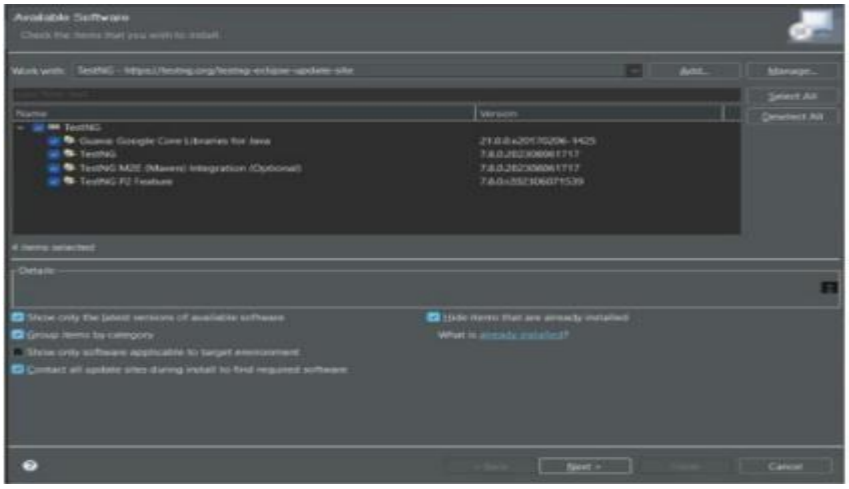
6. Now copy The URL https://testng.org/testng-eclipse-update-site And Past The Work With Option Bar.



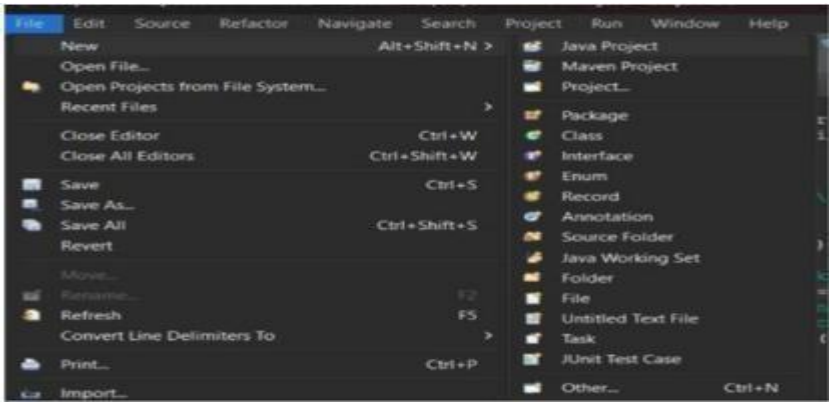
7. Then Click Add Button And Past The URL https://testng.org/testngeclipseupdate-site and Give The Name (TestNG). And Click the Add Button.



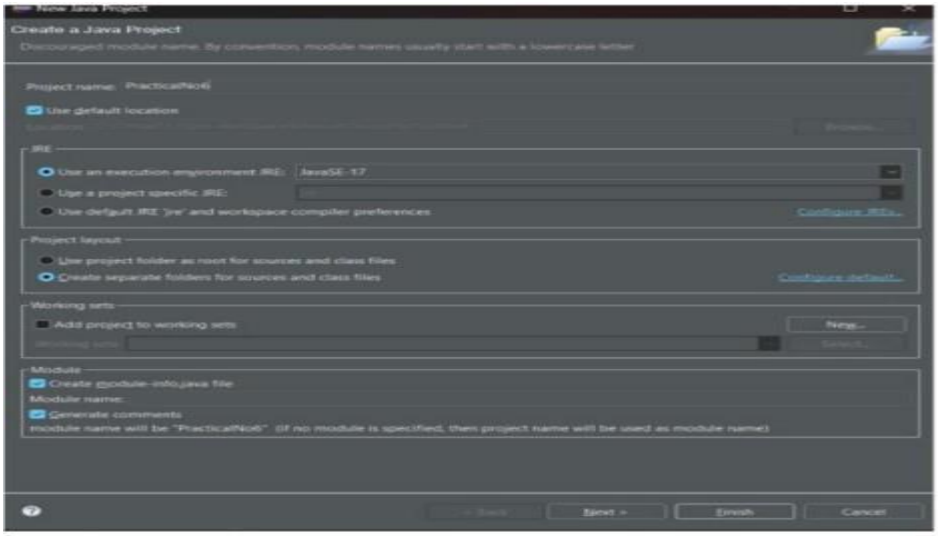
8. Add All plugins And Click on the next button.



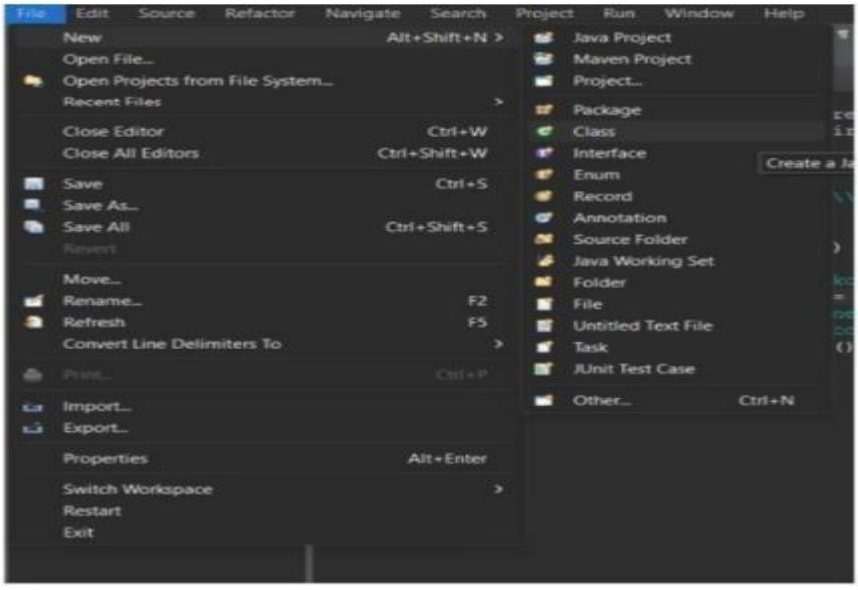
Step 4: Now open the Eclipse Create a new java project.



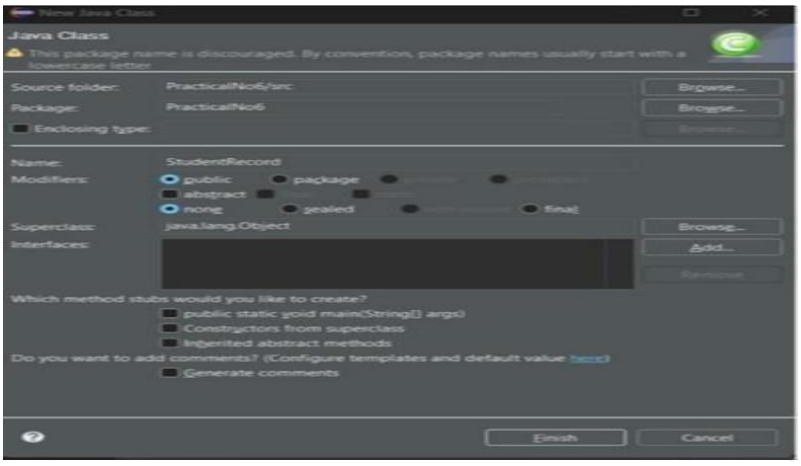
Step 5: Now Give the Project Name .



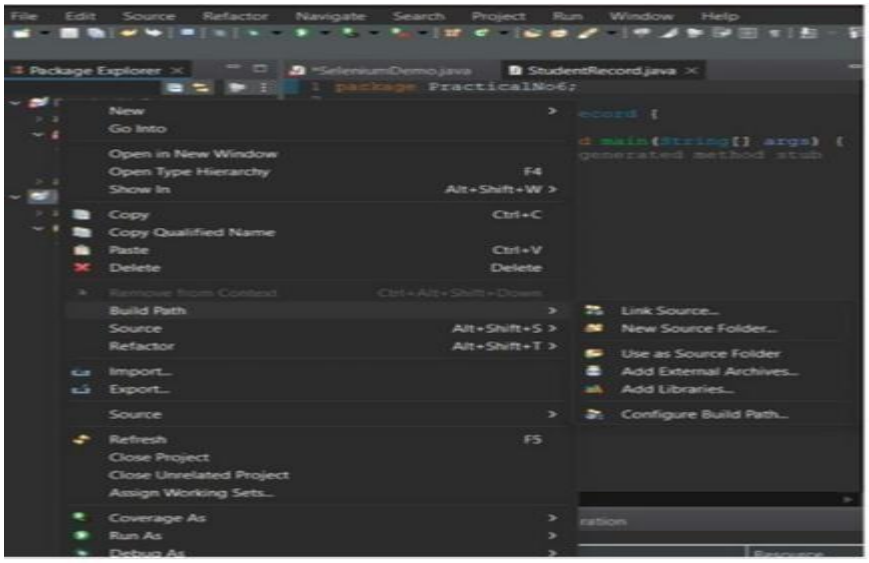
Step 6: Now Create the java Class . So go to the Eclipse IDE Click the File , Select the New , And Click on the Class.



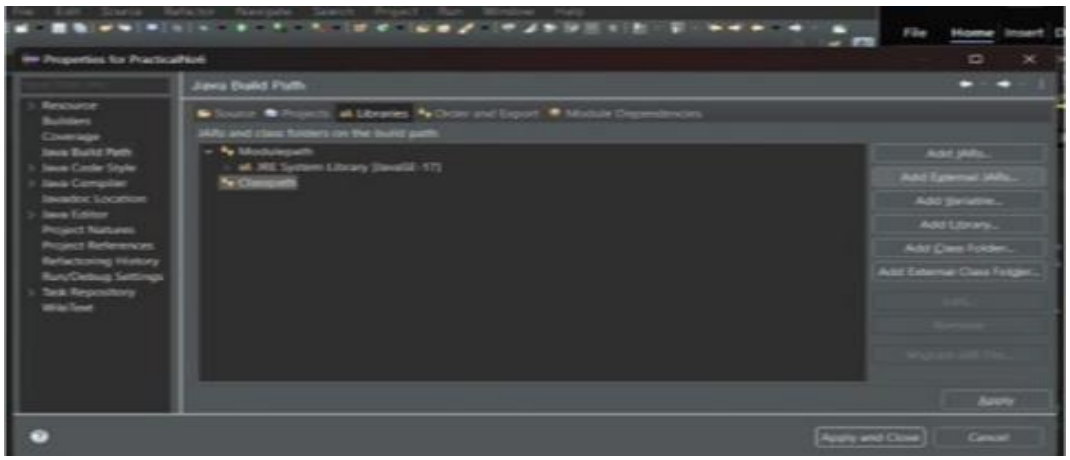
Step 7: Give the class name(StudentRecord) and Click on the Finish button. Select on the public static void main(String[] args).



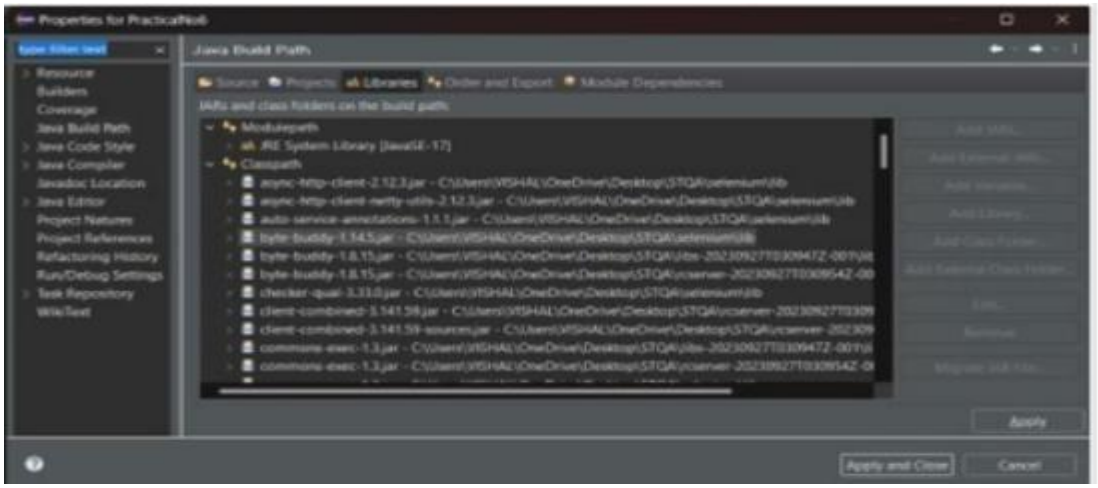
Step 8: Now We need to Some Libraries References. Then click on the file > select The Build Path > Click on the Config Build Path.



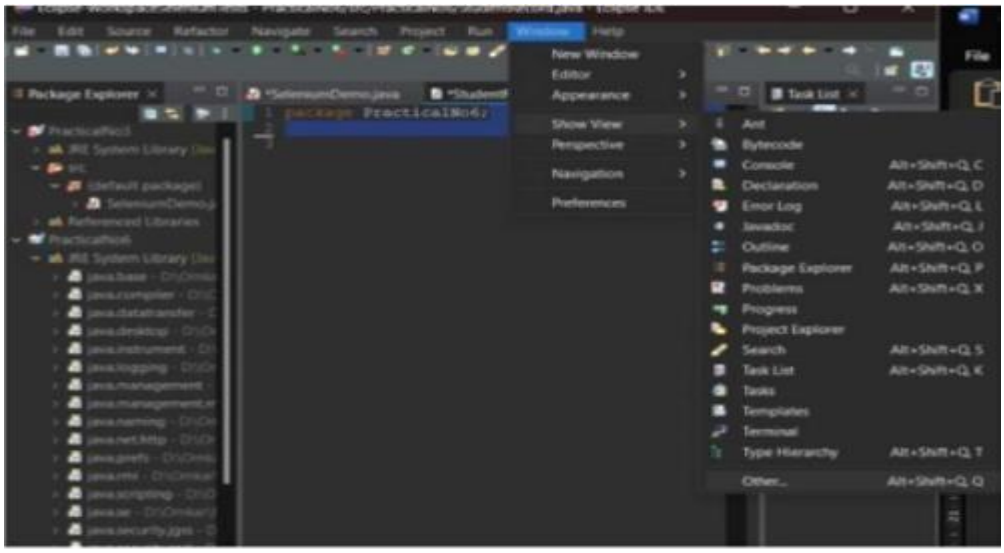
Step 9 : Go to the Libraries section and select the classpath , and click on the Add External JARS.



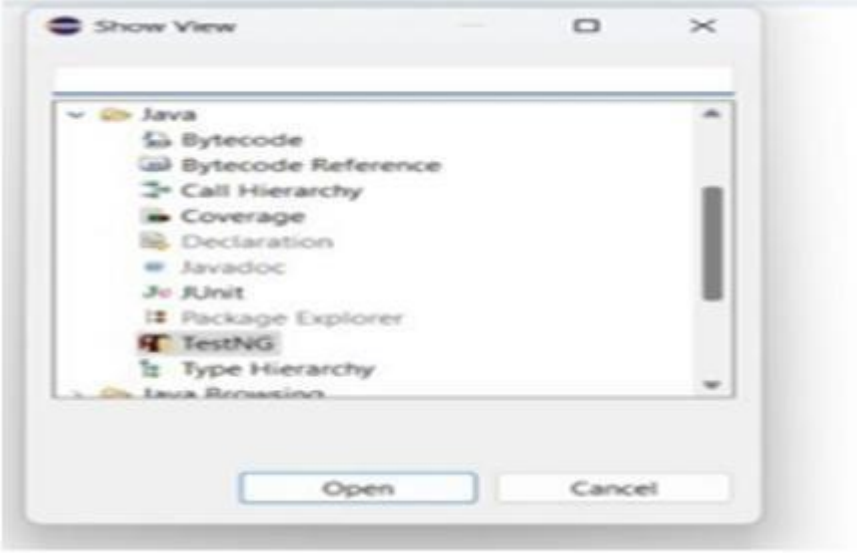
Step 10 : Then Add the .Jar files and click on the Apply and Close.



Step 11 : Then Add The TestNG . So Go to the Windows option and click the Show view > Other .



Step 12: Go to Other and Click the Java Folder and select the TestNG.



Step 13: Then Run The Code.

Step 14: Then Copy the path of the Excel Workbook.

Code:

```
import java.io.File;  import
jxl.Sheet; import jxl.Workbook;
import jxl.write.Label; import
jxl.write.WritableSheet; import
jxl.write.WritableWorkbook;
public class Result {
   public static void main(String[] args) throws Exception {
    Workbook wb = Workbook.getWorkbook(new File("input.xls"));
    Sheet s = wb.getSheet(0);
     WritableWorkbookwwb =
           Workbook.createWorkbook(new File("output.xls"));
    WritableSheet ws = wwb.createSheet("Result", 0);
String[[] a = new String[s.getRows()][s.getColumns()];  for
(int i = 0; i < s.getRows(); i++) {  for (int j = 0; j <
s.getColumns(); j++) {  a[i][j] = s.getCell(j,
i).getContents();  Label l2 = new Label(j, i, a[i][j]);
ws.addCell(l2);
          }
        }
 Label l1 = new Label(6, 0, "Result");
ws.addCell(l1);            for (int i = 1; i <
s.getRows(); i++) {
 for (int j = 2; j < s.getColumns(); j++) {
a[i][j] = s.getCell(j, i).getContents();
int x = Integer.parseInt(a[i][j]);
if (x > 35) {
           Label result = new Label(6, i, "pass");
ws.addCell(result);
           } else {

             Label result = new Label(6, i, "fail");
ws.addCell(result);

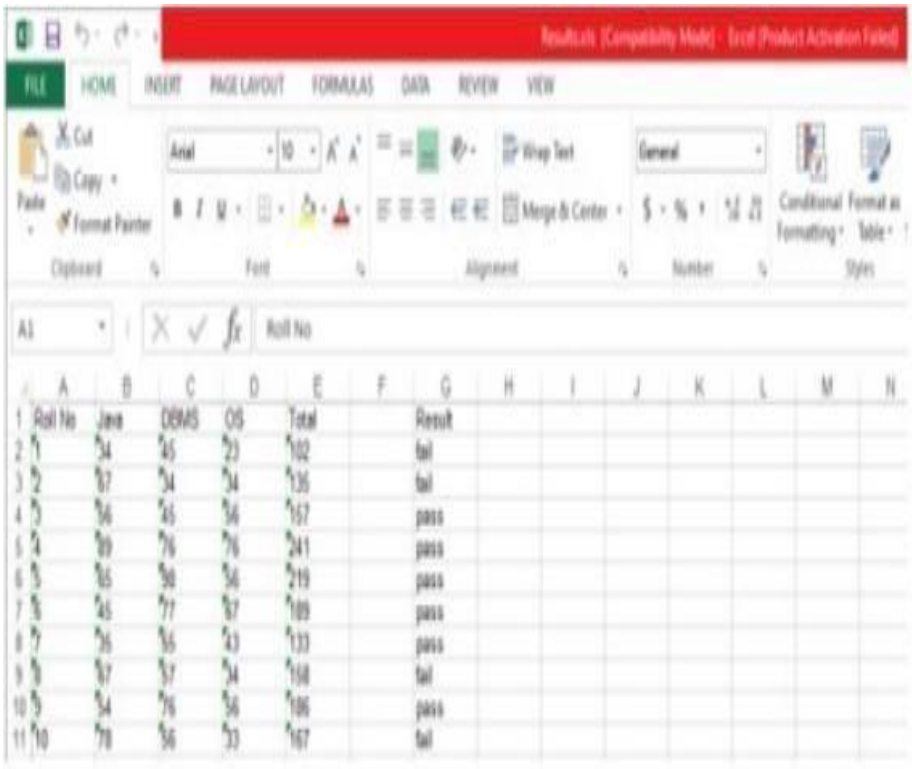
break;            }
          }
        }

       System.out.println("Records successfully updated");
```

```
        // Save the Excel file
wwb.write();

        // Close files
wwb.close();
wb.close();
    }
}
```

OUTPUT:



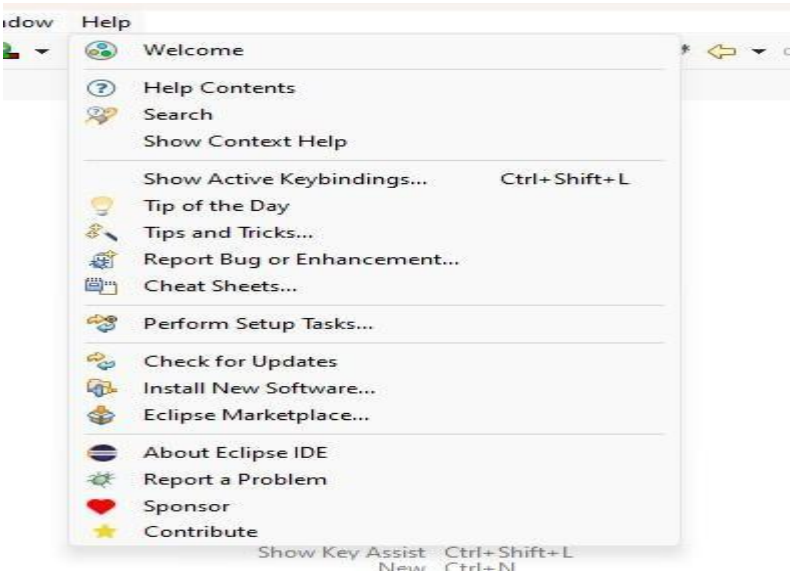
	A	B	C	D	E	F	G
1	Roll No	Java	DBMS	OS	Total		Result
2	1	34	46	23	102		fail
3	2	57	34	34	135		fail
4	3	56	45	56	157		pass
5	4	89	76	76	241		pass
6	5	65	98	56	219		pass
7	6	45	77	67	189		pass
8	7	26	56	43	133		pass
9	8	67	57	34	158		fail
10	9	54	76	56	186		pass
11	10	78	56	33	167		fail

PRACTICAL NO. 6

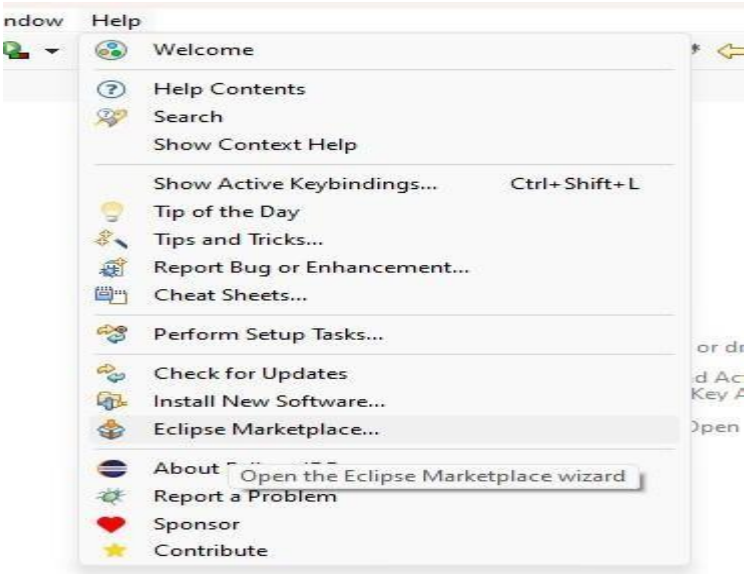
WRITE AND TEST A PROGRAM TO SELECT THE NUMBER OF STUDENTS WHO HAVE SCORED MORE THAN 60 IN ANY ONE SUBJECT (OR ALL SUBJECTS).

Step 1: Go to Google and Search Junit-4.10.jar. Use URL: http://www.java2s.com/code/Jar/j/Downloadjunit410jar.htm Step 2: Then Download Jxl2.6.jar file. Use URL: http://www.java2s.com/Code/Jar/j/Downloadjxl26jar.htm Step 3: We need TestNG. Then open Eclipse IDE.

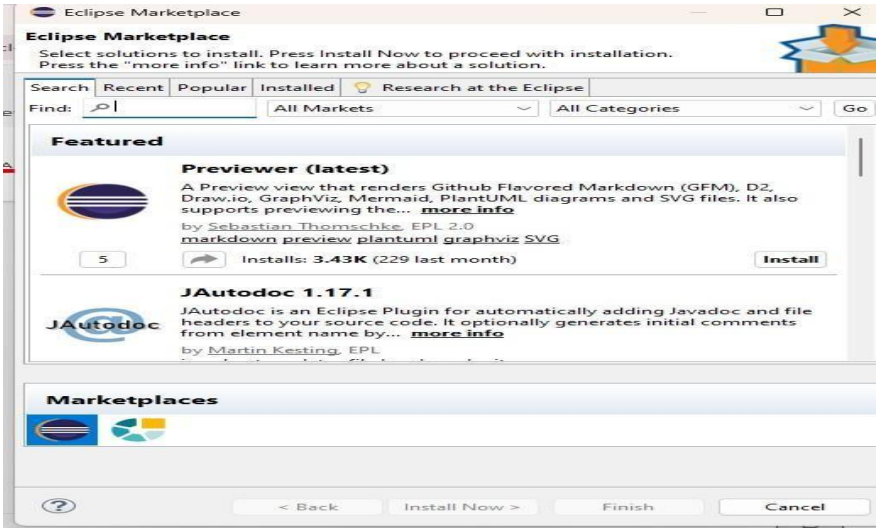
1. On the menu bar, click Help.



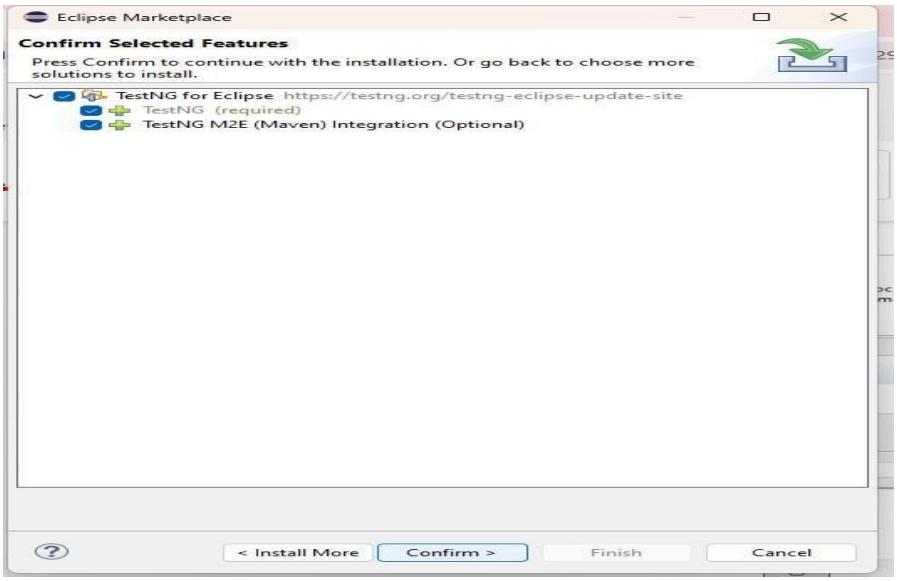
2. Then Select Eclipse Marketplace.



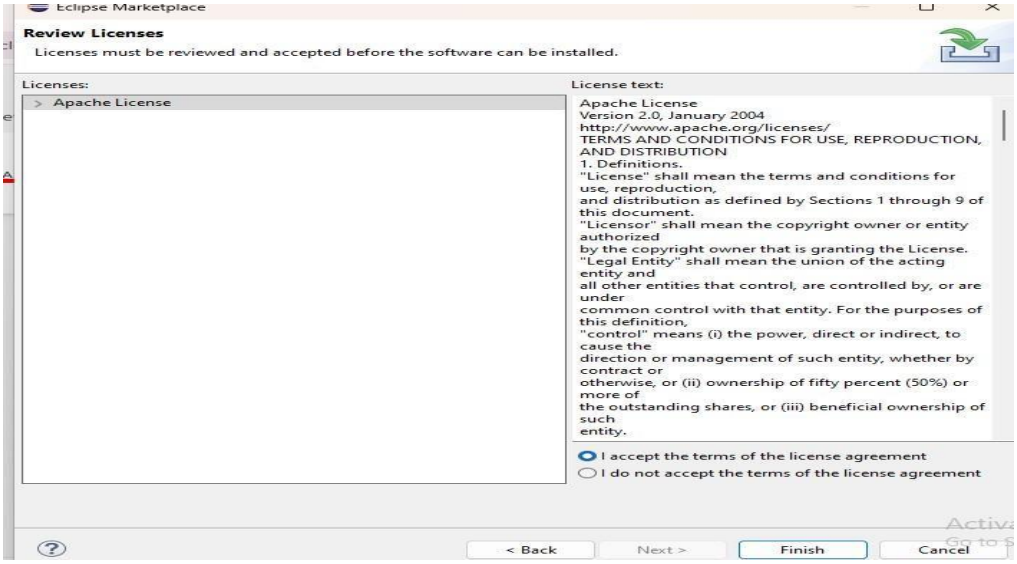
3. Search the TestNG using the Find option and click on installation button



4. Confirm the installation process and click on Confirm button.

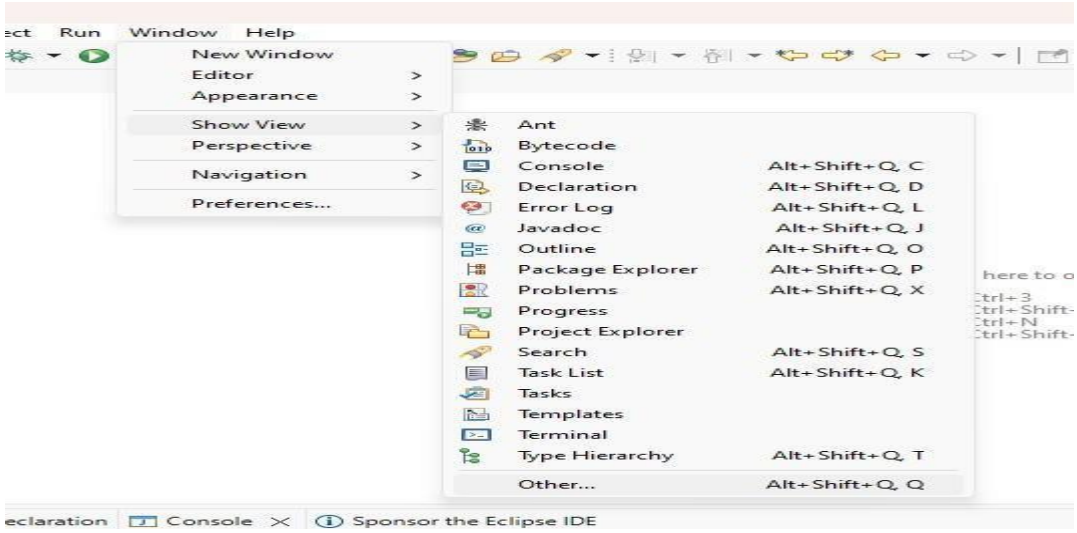


5. Accept the license agreement and select the radio button “I accept the terms of the license agreement”. Click on Finish.

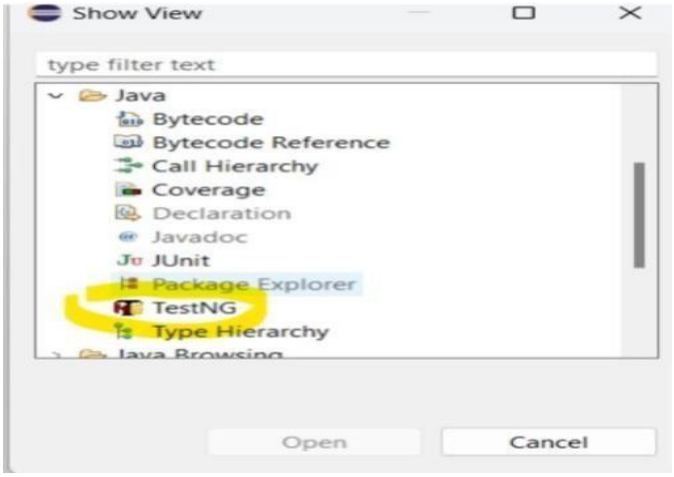


6. After accepting the license agreement, wait for installation.

7. Verify if the installation is done properly. After the restart, verify if TestNG for Eclipse was indeed successfully installed. Click Window > Show View > Other.



8. Then open the Java directory and see if TestNG is included. Select TestNG and click Open button. Now It's Done.



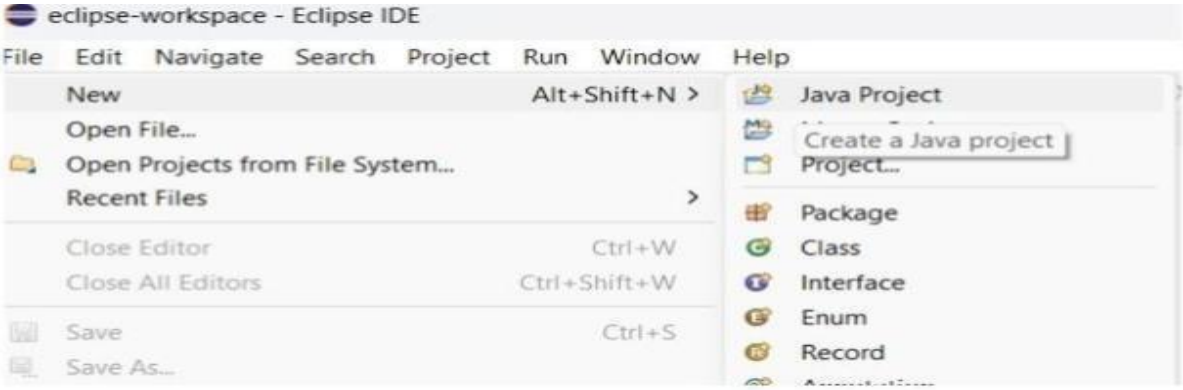
Step 4: Create an input file in Excel File with 10 records. Give the Student Name of this excel file (Student.xls).

Roll No	JAVA	DBMS	OS	Total
1	34	45	23	102
2	67	34	34	135
3	56	45	56	157
4	89	76	76	241
5	65	98	56	219
6	45	77	67	189
7	35	55	43	133
8	67	57	34	158
9	54	76	56	186
10	78	56	33	167

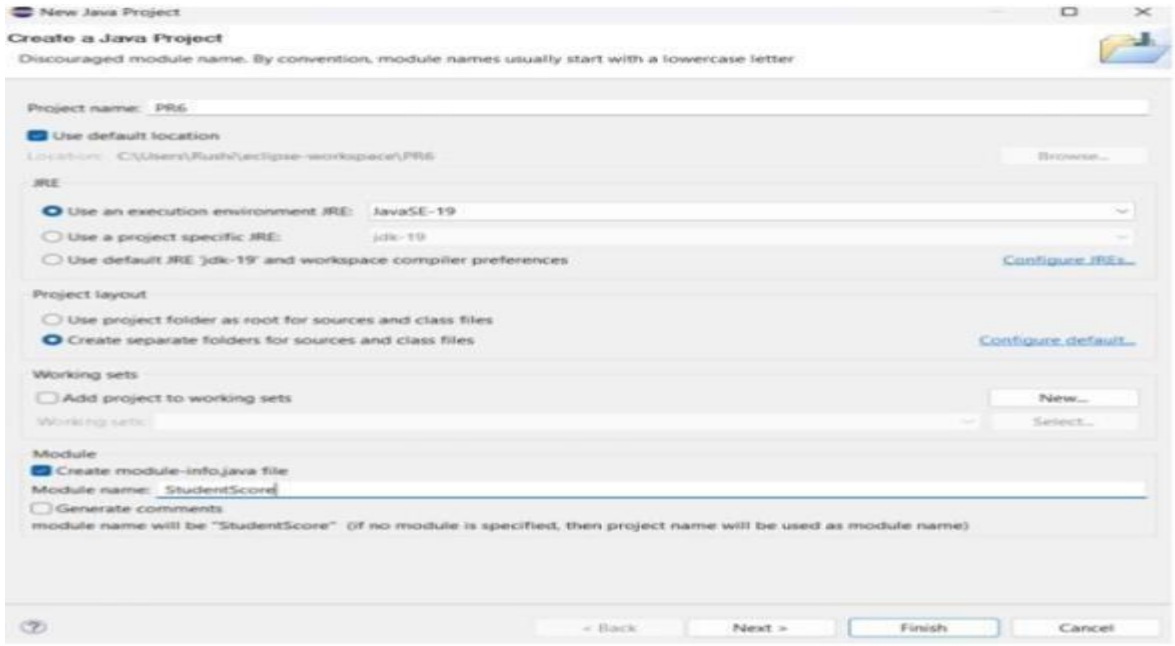
Step 5: Create an Empty Excel Output file. Save this name (Result.xsl).

	A	B	C	D	E
1	Roll No	JAVA	DBMS	OS	Total
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					

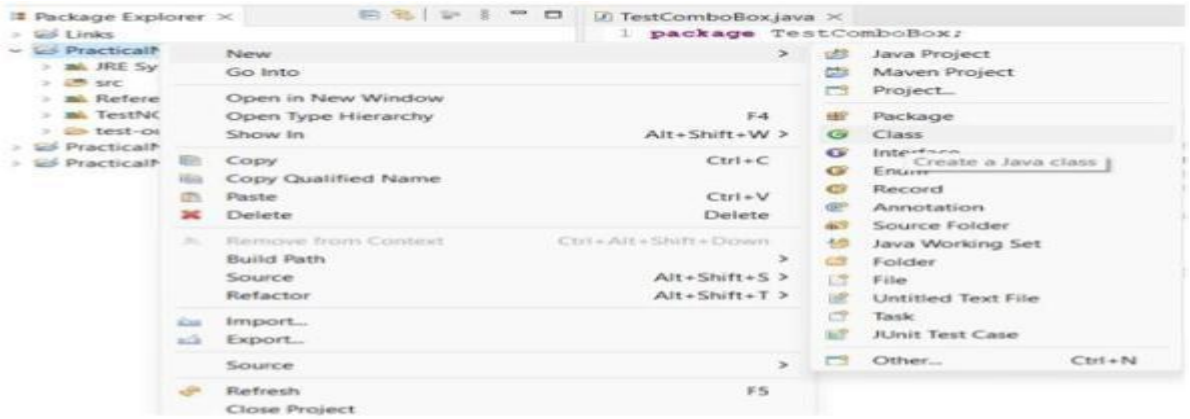
Step 6: Open the Eclipse IDE and go to File and then select the New option, then click Java Project.



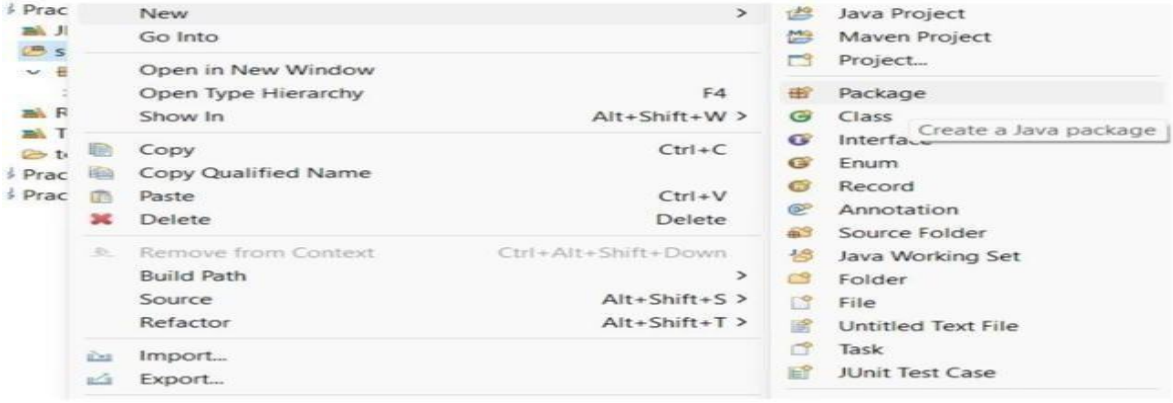
Step 7: Give the Project Name and then click Next Button.



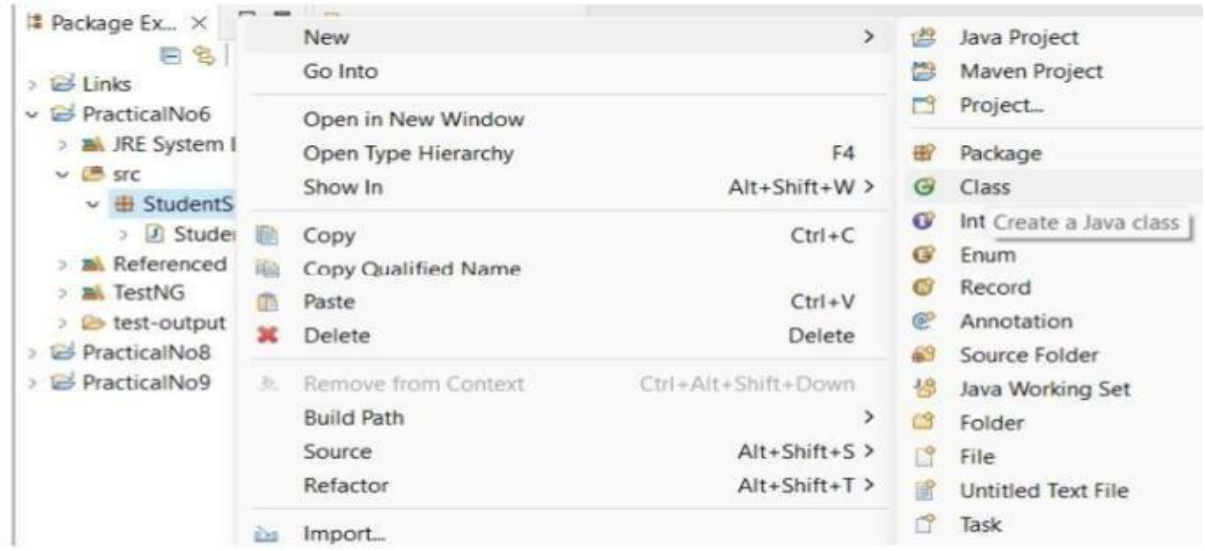
Step 8: Go to the Left Side and click on the project, then click on the New button and select Class



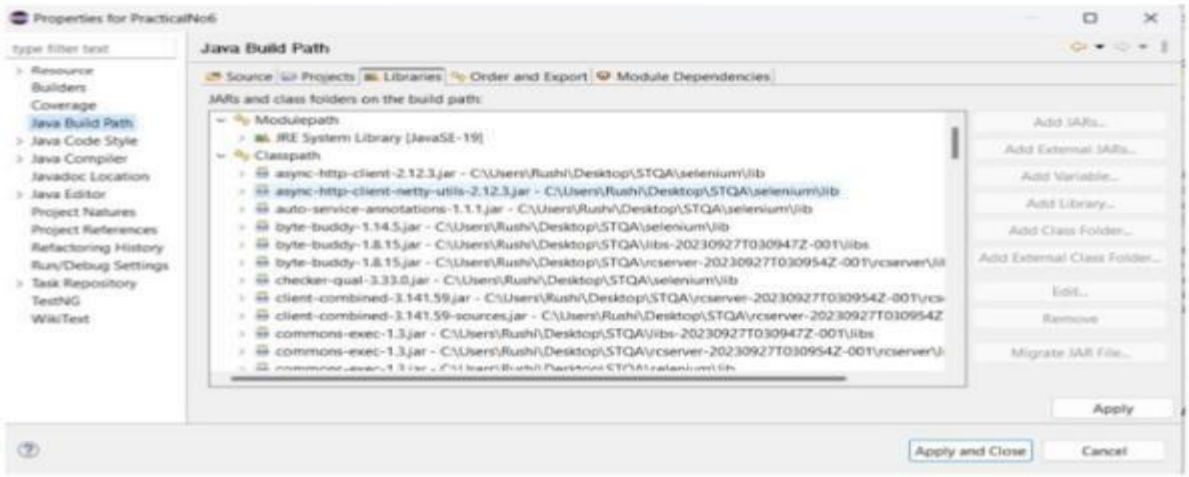
Step 9: Click Right Button on Project Src and create a package.



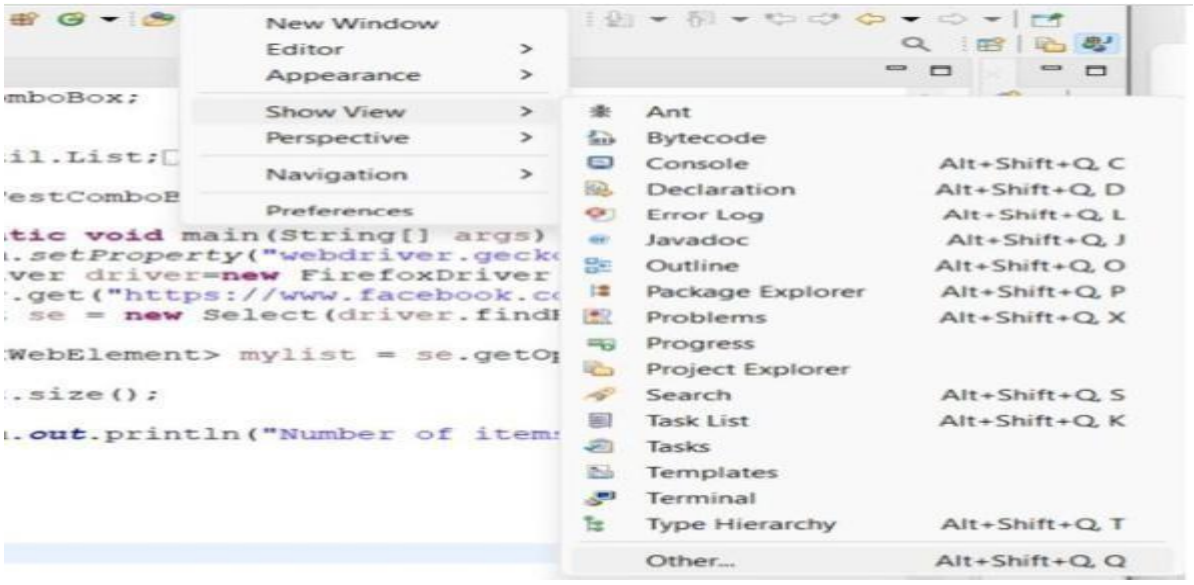
Step 10: Then Right click on the package and Add a Class and Class Name Is (StudentScore) in both Source Folder and Name.

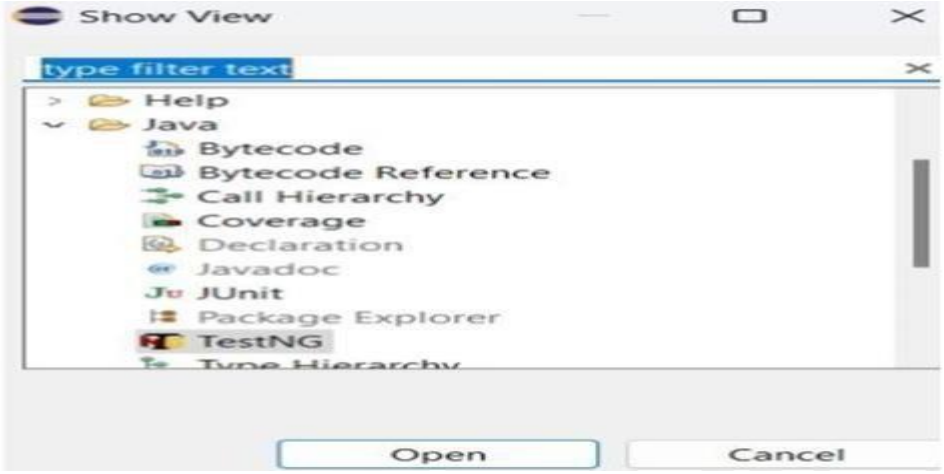


Step 11: Add referenced Libraries.



Step 12: Go to Windows Option in Eclipse, then select Show View. Go to the Other Option, click on Java Folder and select the TestNG.





Step 13: Type the code.

```
package StudentScore; import
jxl.Sheet; import jxl.Workbook;
import jxl.write.*; import
java.io.*; import
org.testng.annotations.Test;


@Test public class
StudentScore {
 public void testImportexport1() throws Exception {
FileInputStream fi = new

FileInputStream("C:\Users\Rushi\Desktop\Student.xls");
 Workbook w = Workbook.getWorkbook(fi);
 Sheet s = w.getSheet(0);
 String a[][] = new String[s.getRows()][s.getColumns()];
 FileOutputStream fo = new

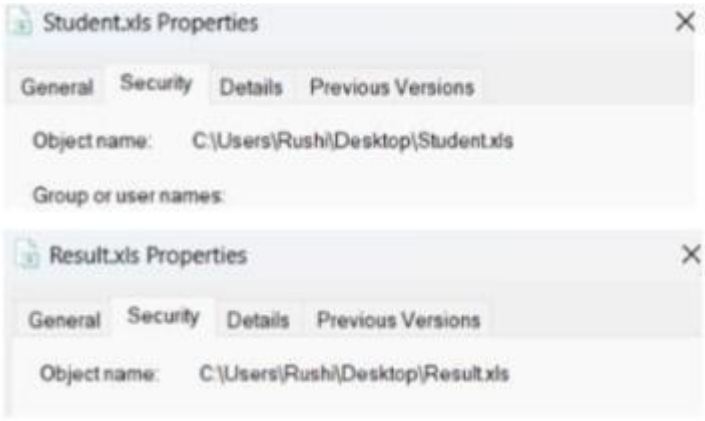
FileOutputStream("C:\Users\Rushi\Desktop\Result.xls");
WritableWorkbook wwb = Workbook.createWorkbook(fo);
WritableSheet ws = wwb.createSheet("result", 0);  int c = 0;


for (int i = 0; i < s.getRows(); i++) {
for (int j = 0; j < s.getColumns(); j++) {
if (i >= 1) {  String b = new String();  b
= s.getCell(3, i).getContents();  int x =
Integer.parseInt(b);  if (x < 60) {  c++;
break;  }
 }


a[i][j] = s.getCell(j, i).getContents();
Label l2 = new Label(j, i - c, a[i][j]);
ws.addCell(l2);
 }
 }

System.out.println("Students scored less than 60: " + c);
wwb.write();  wwb.close();
 } }
```

Step 14: Then Copy the Created Both Excel File And Paste in Code.



Step 15: Run The Code.

OUTPUT:

	A	B	C	D	E
1	Roll No	Java	DBMS	OS	Total
2	4	89	76	76	241
3	6	45	77	67	189

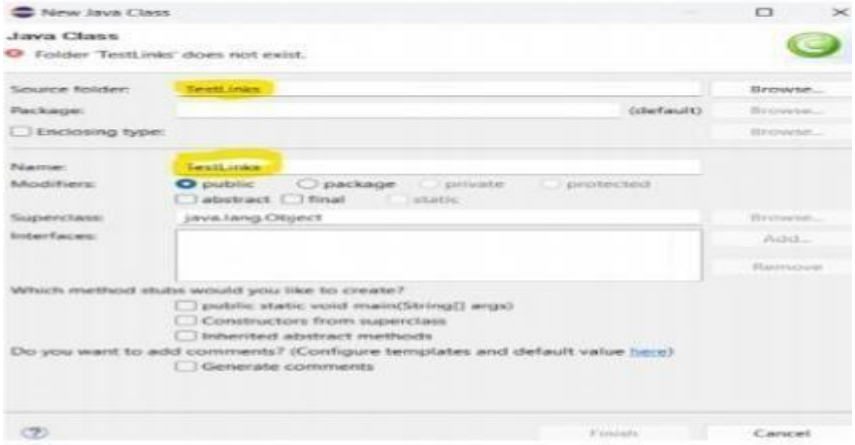
PRACTICAL NO. 7

WRITE AND TEST A PROGRAM TO PROVIDE TOTAL NUMBER OF OBJECTS AVAILABLE IN THE WEB PAGE.

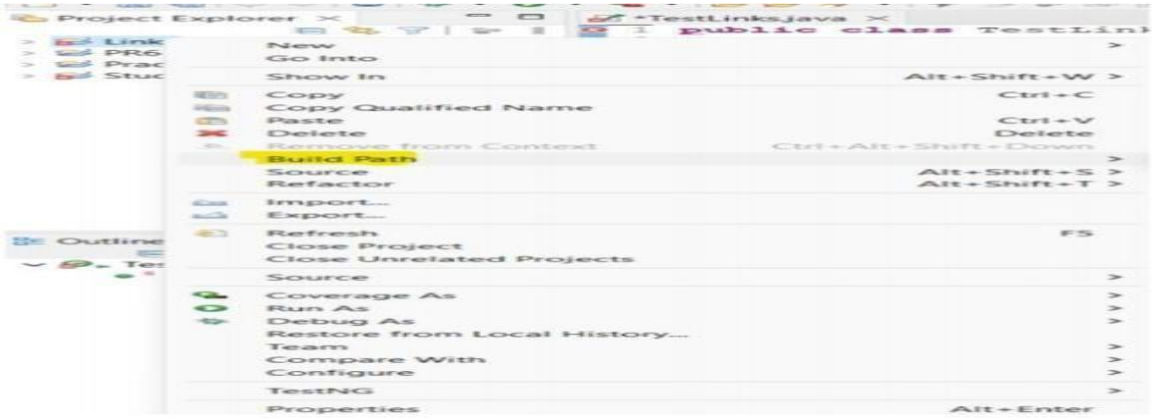
Step 1: Download Selenium Server from Server Grid. Step
2: Open The Eclipse IDE.
Step 3: Create a JAVA Project. Give the Name Links.



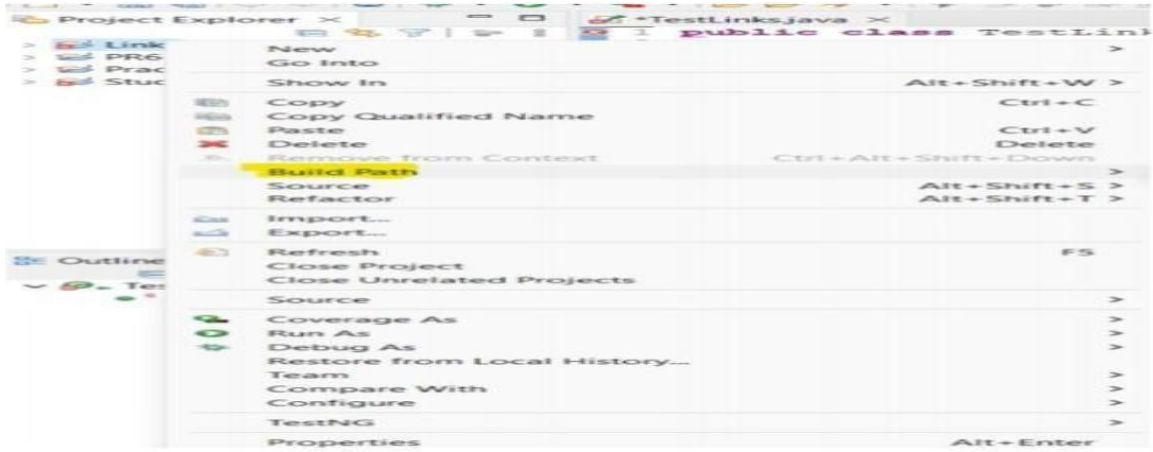
Step 4: Then Add a Class and Class Name Is (TestLinks) in Both Source Folder and Name.



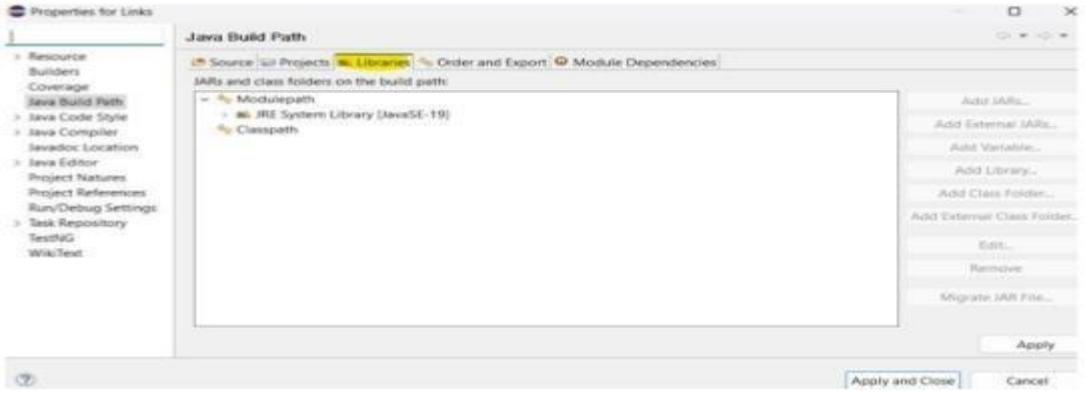
Step 5: Then Select the public static void main(String[] args) and click Finish button.



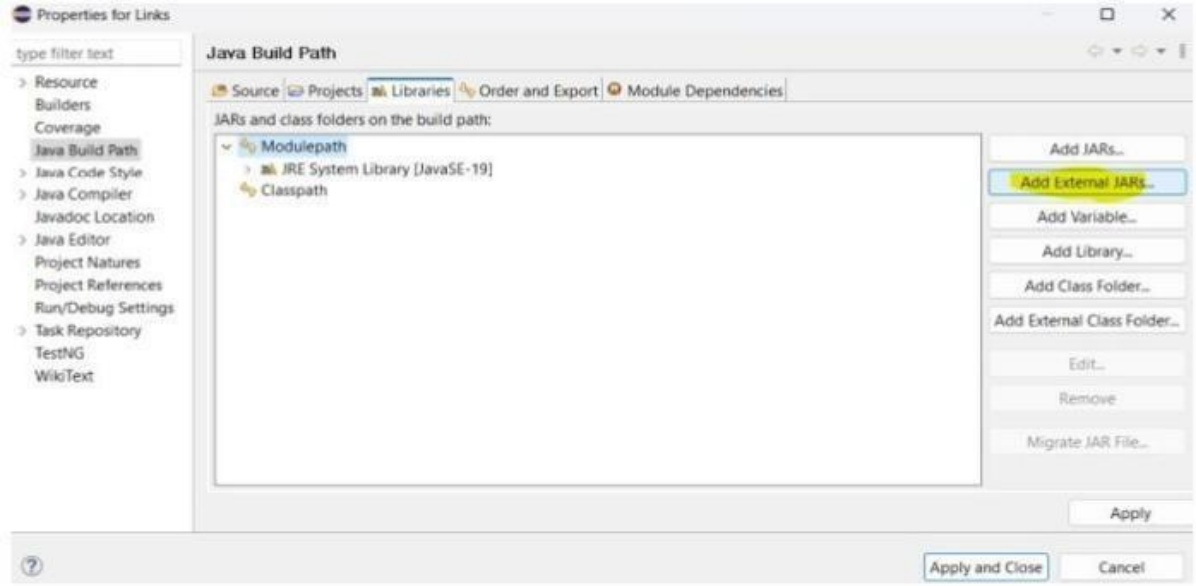
Step 6: Add All The Basics Selenium jars. Go to the left side Package Explorer, click the right button on the Project Name, choose a JAVA Build Path, then select Configuration Build Path.



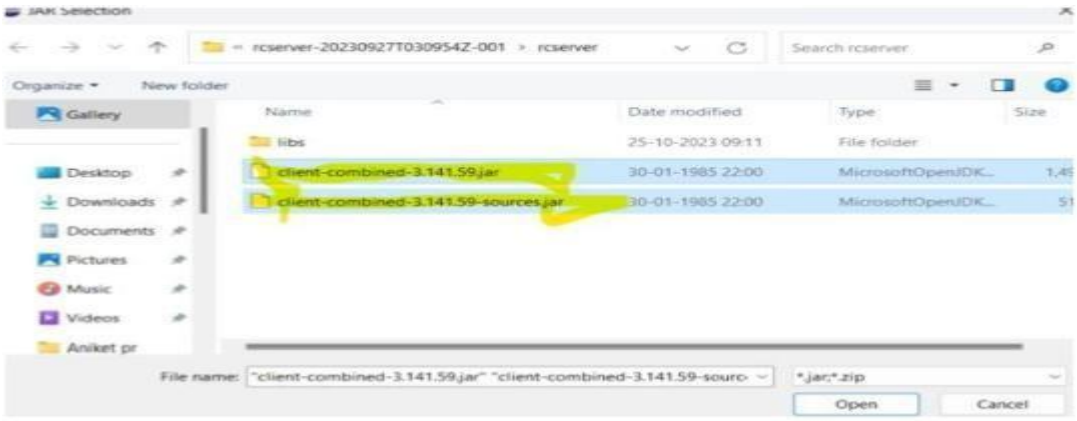
Step 7: Go to the Libraries.



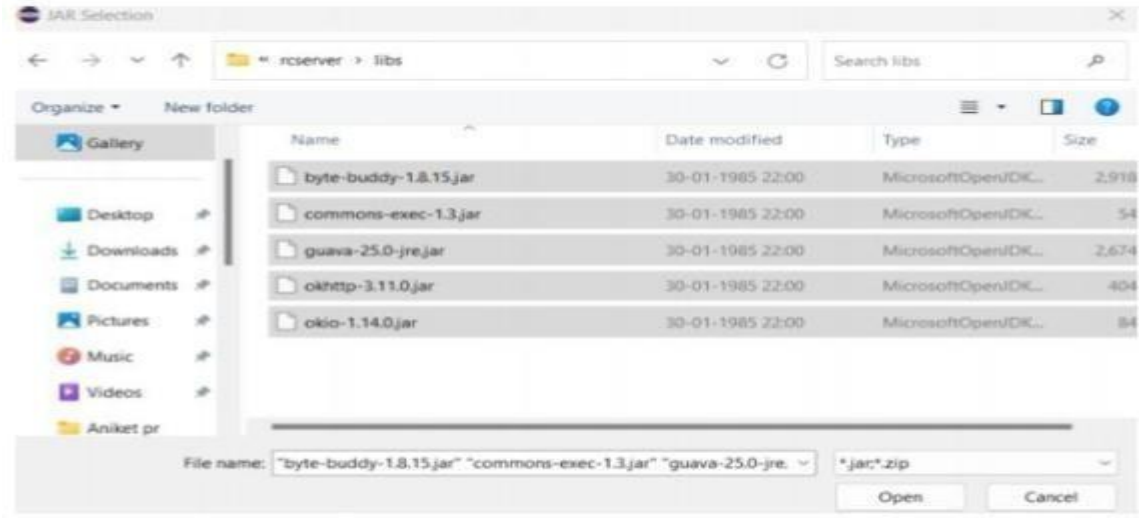
Step 8: Go to Right Side and click on the Add External JARS.



Step 9: Then Select rcserver folder, select all .jar files, then click the Open button.



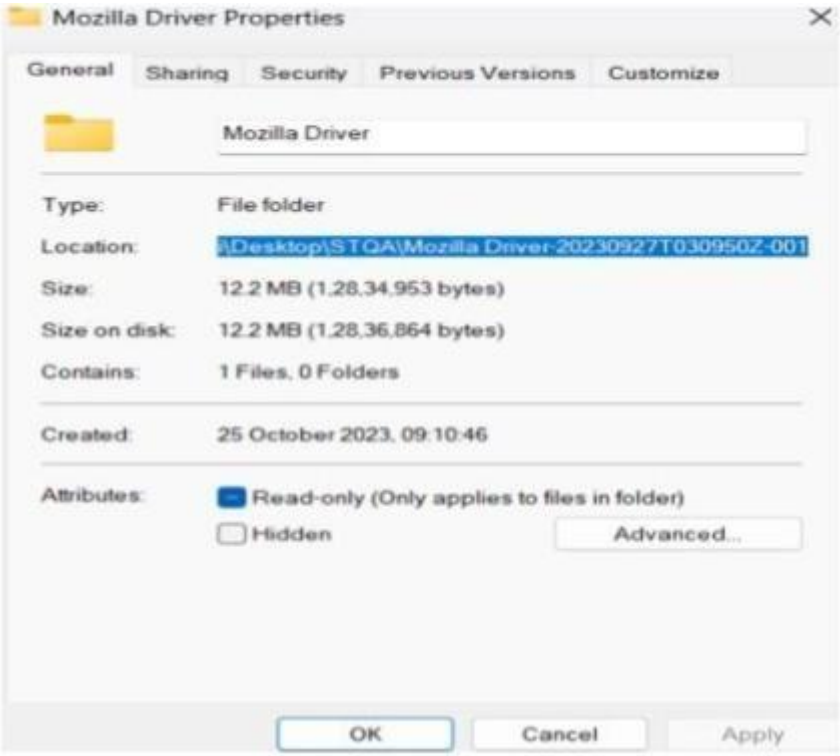
Step 10: Select Again External Jar option, then select rcserver, click lib folder and select all Jars files. Click on Open button. Then Apply and Close.



Step 11: Then Type Full Code. import java.util.List; import org.openqa.selenium.By; import org.openqa.selenium.WebDriver; import org.openqa.selenium.WebElement; import org.openqa.selenium.firefox.FirefoxDriver;

```
public class TestLinks {
 public static void main(String[] args) {
System.setProperty("webdriver.gecko.driver","yourLocationToGeckoDriver\geckodriver.exe
");
 WebDriver driver = new FirefoxDriver();  driver.get("https://www.gmail.com");
 List<WebElement> mylist = driver.findElements(By.xpath("//a"));
 System.out.println("Number of links =" + mylist.size());
 }
}
```

Step 12: Give the GeckoDriver path. Go to the file folder. Open STQA folder, go to the Mozilla Driver, then go to the GEKODRIVER properties and copy the path.



Step 13: Then Run The Project.

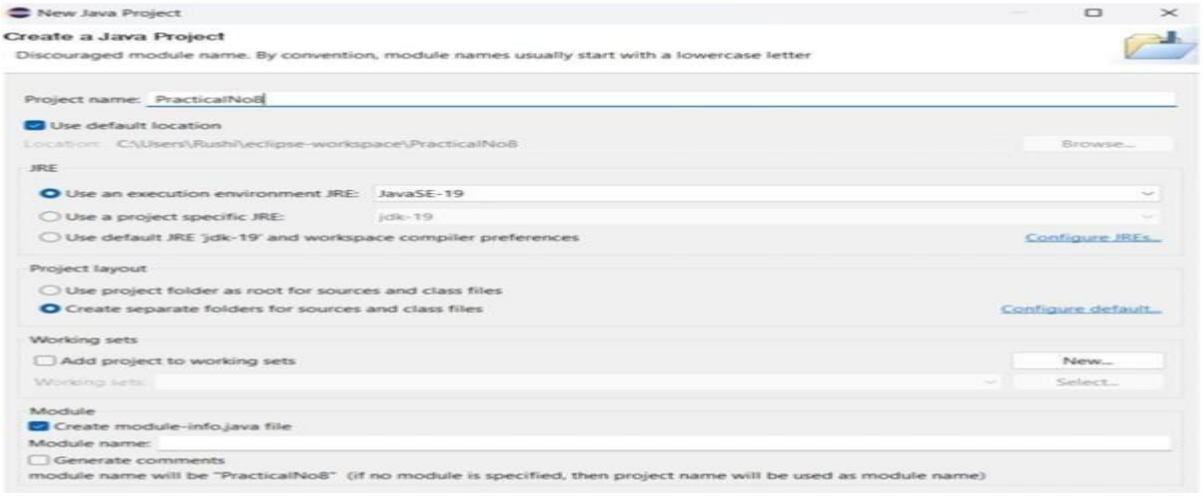
```
Oct 30, 2023 12:47:59 PM org.openqa.selenium.remote.Protocol
INFO: Detected dialect: W3C
1698650303921   addons.xpi      ERROR   System addon update
Number of links =4
```

PRACTICAL NO. 8

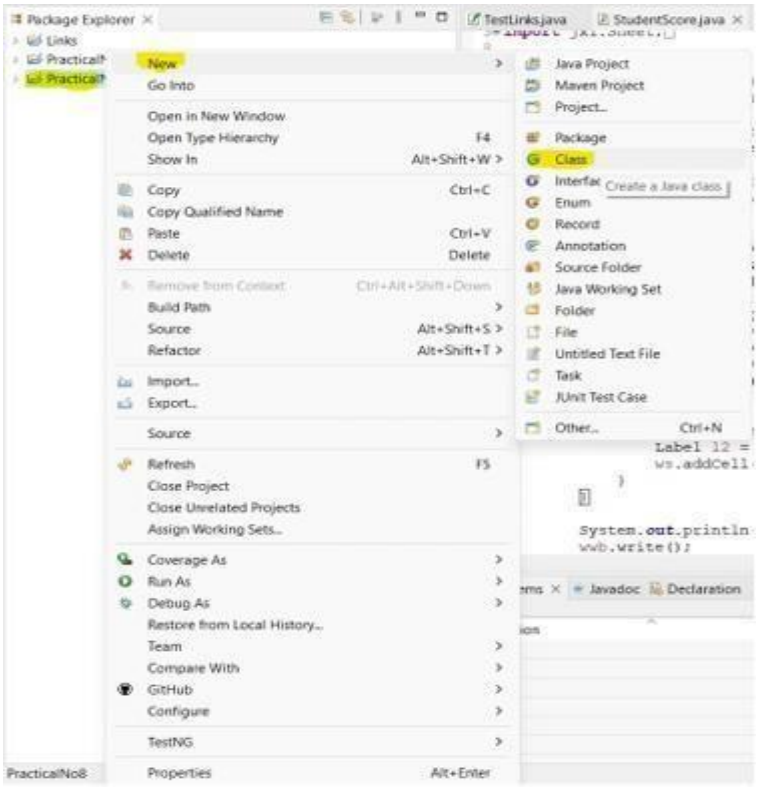
WRITE AND TEST A PROGRAM TO GET THE NUMBER OF ITEMS IN A LIST / COMBO BOX.

Step 1: Go to the Eclipse IDE.

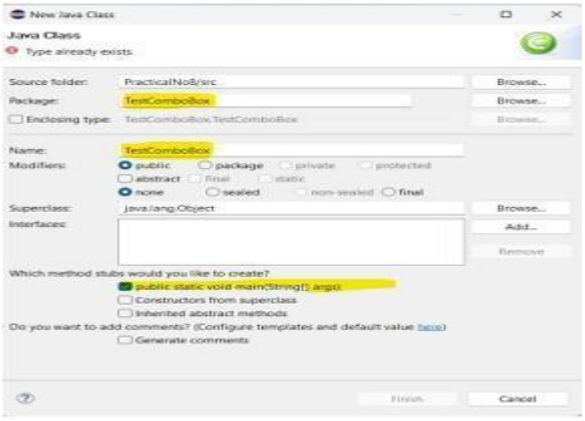
Step 2: Create a JAVA Project. Give the name on the project.



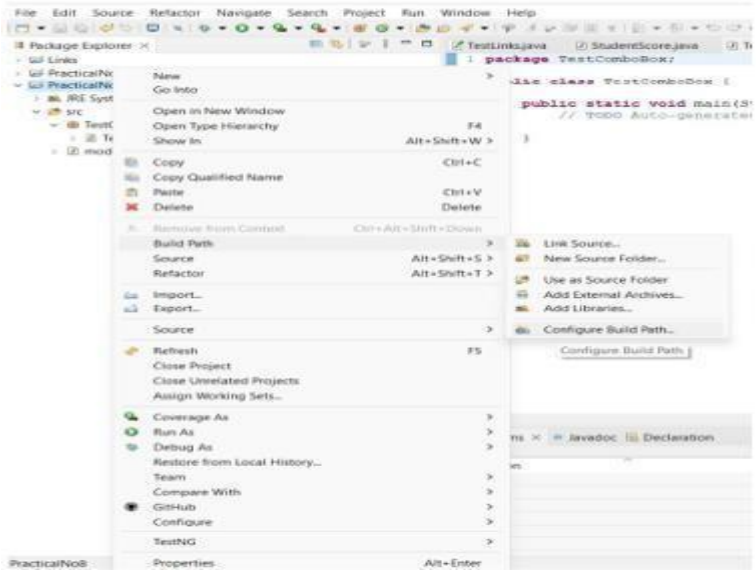
Step 3: Go to the Left Side Bar and Click the Right button on the project and create a new Class.



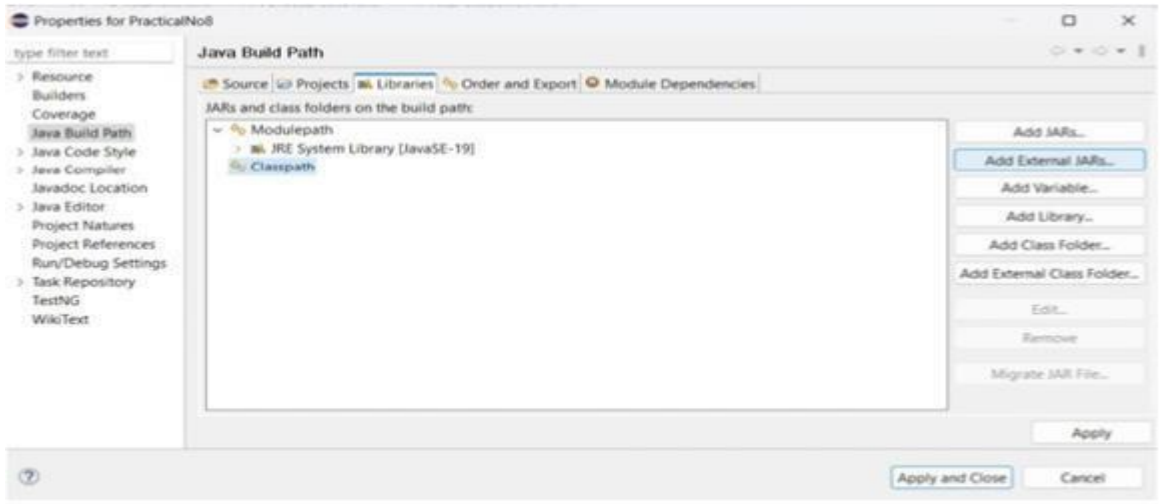
Step 4: Then Give the Class Name and Select public static void main(String[] args). Then click on Finish button.



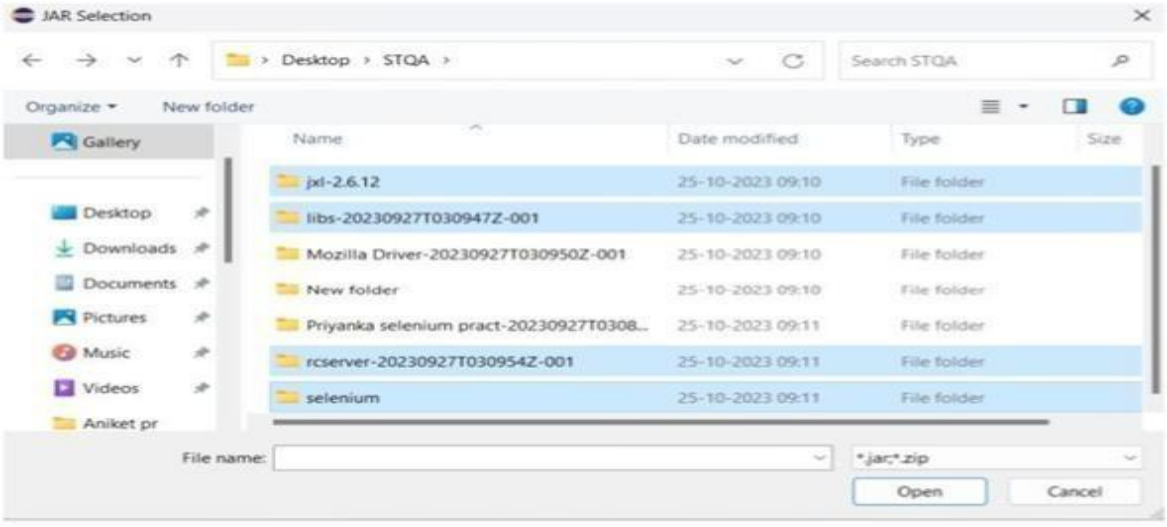
Step 5: Then go again Right Side Click the Right Button On Project Name, click on the Build Path and click on the Configure Build Path.



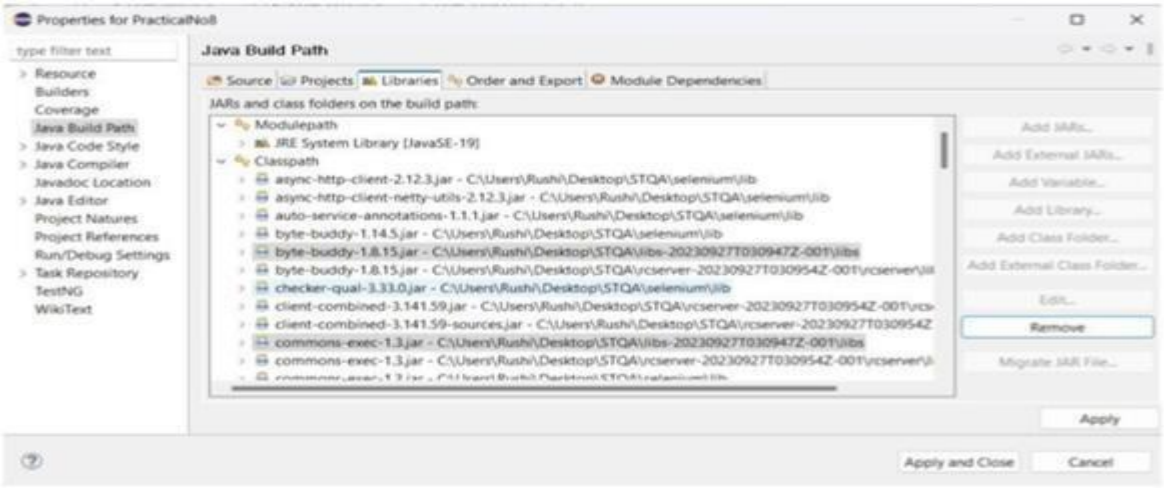
Step 6: Then Go to Libraries and select the Classpath and Click on the Add External Jars.



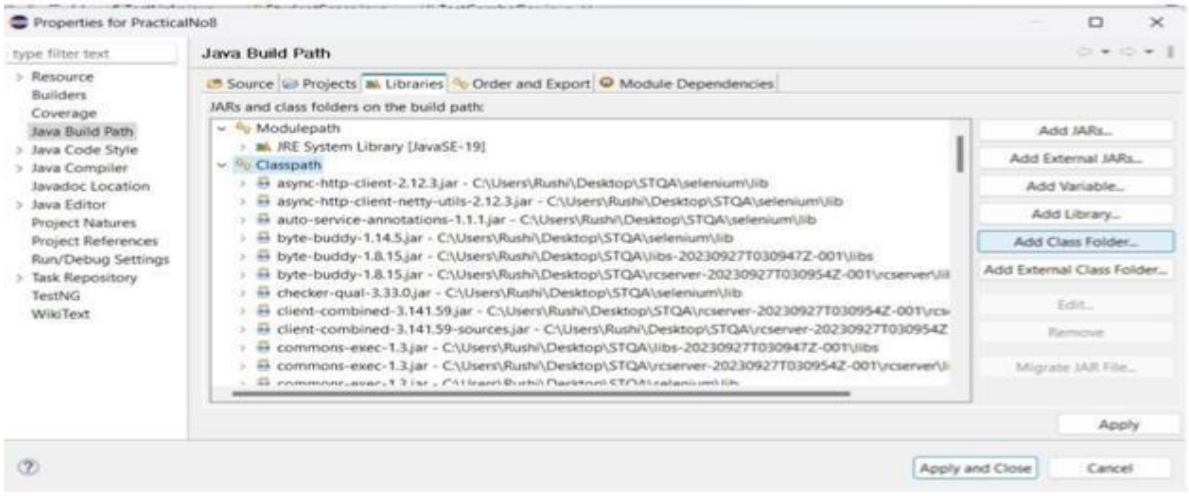
Step 7: Add All Jar files. Go to the rcserver, selenium, lib folders and select all the .jar files.



Step 8: Add All jar files.



Step 9: Add the Jxl.4.2.0 folder. Then go to classpath and click on Add External Class Folder option and add the Jxl folder. Then apply and close.



Step 10: Type code in the project.

```
package TestComboBox;
```

```
import java.util.*; import
org.openqa.selenium.By; import
org.openqa.selenium.WebDriver; import
org.openqa.selenium.WebElement; import
org.openqa.selenium.firefox.FirefoxDriver; import
org.openqa.selenium.support.ui.Select; public
class TestComboBox

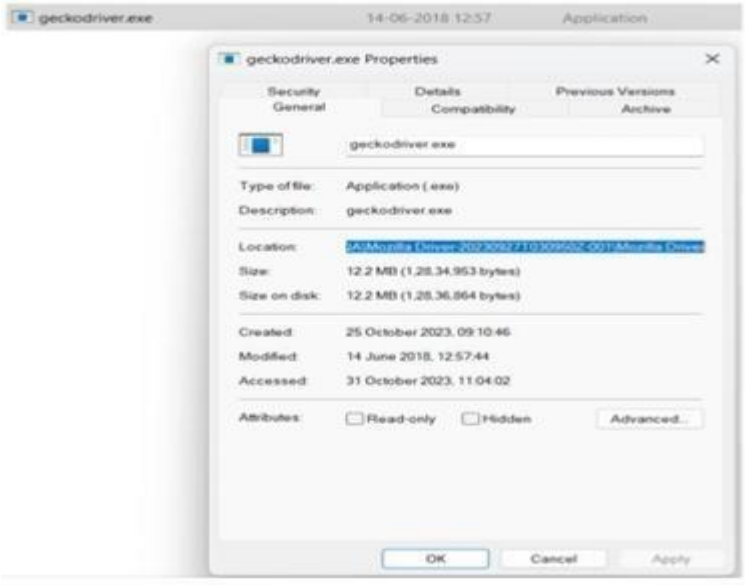
{  public static void main(String
args[])

{
 System.setProperty("webdriver.gecko.driver","E:\soft\software

testing\Mozilla Driver\geckodriver.exe");
WebDriver driver = new FirefoxDriver();
driver.get("https://www.facebook.com"); Select
se = new

Select(driver.findElement(By.xpath("//Select[@id='month']")));
List<WebElement> mylist = se.getOptions();  mylist.size();
System.out.println("Number of items = "+mylist.size());
 }
}
```

Step 11: Then need to GeckoDriver path. Open Mozilla Driver folder and copy geckodriver.exe path.



Step 12: Then Add GeckoDriver Path in code.



Step 13: Then Run the code.

Code:

package Prac08; import java.util.List; import org.openqa.selenium.By; import org.openqa.selenium.WebDriver; import org.openqa.selenium.WebElement; import org.openqa.selenium.firefox.FirefoxDriver; import org.openqa.selenium.support.ui.Select; public class P08 {
public static void main(String[] args) {
System.setProperty("webdriver.gecko.driver","D:\STQA\Mozilla Driver-20230927T030950Z-001\Mozilla Driver\geckodriver.exe");
WebDriver driver=new FirefoxDriver();
driver.get("https://www.facebook.com");
Select se = new Select(driver.findElement(By.xpath("//Select[@id='month']")));
List<WebElement> mylist = se.getOptions(); mylist.size();
System.out.println("Number of items = "+mylist.size());
}
}

OUTPUT:



PRACTICAL NO. 9

WRITE AND TEST A PROGRAM TO COUNT THE NUMBER OF CHECK BOXES ON THE PAGE, CHECKED AND UNCHECKED COUNT.

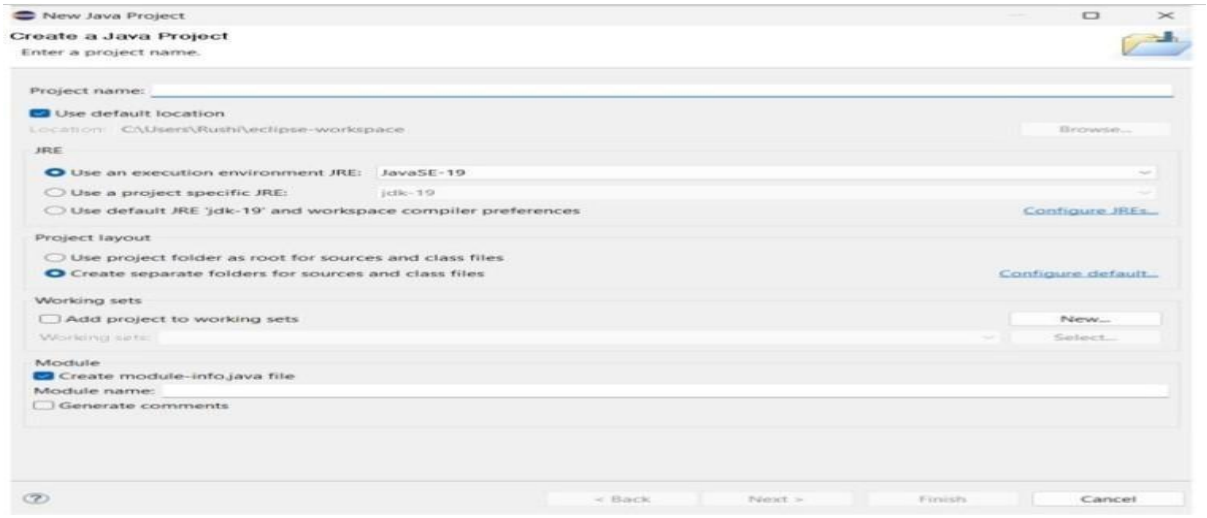
Step 1: Open the Notepad.
Step 2: Create a basic HTML page with some of the checkboxes checked and unchecked.
HTML Code:

```
<!DOCTYPE html>
<html>
<body>
<form>
<h2>Text Input</h2>
First name:<br>
<input type="text" name="firstname"><br>
Last name:<br>
<input type="text" name="lastname">
<br>
<h2>Select Gender</h2>
<input type="radio" name="gender" value="male" checked> Male<br>
<input type="radio" name="gender" value="female"> Female<br>
<input type="radio" name="gender" value="other"> Other<br> <h2>Select
Languages Known</h2>
<input type="checkbox" name="lang" value="Java"> Java<br>
<input type="checkbox" name="lang" value="PHP" checked="checked">

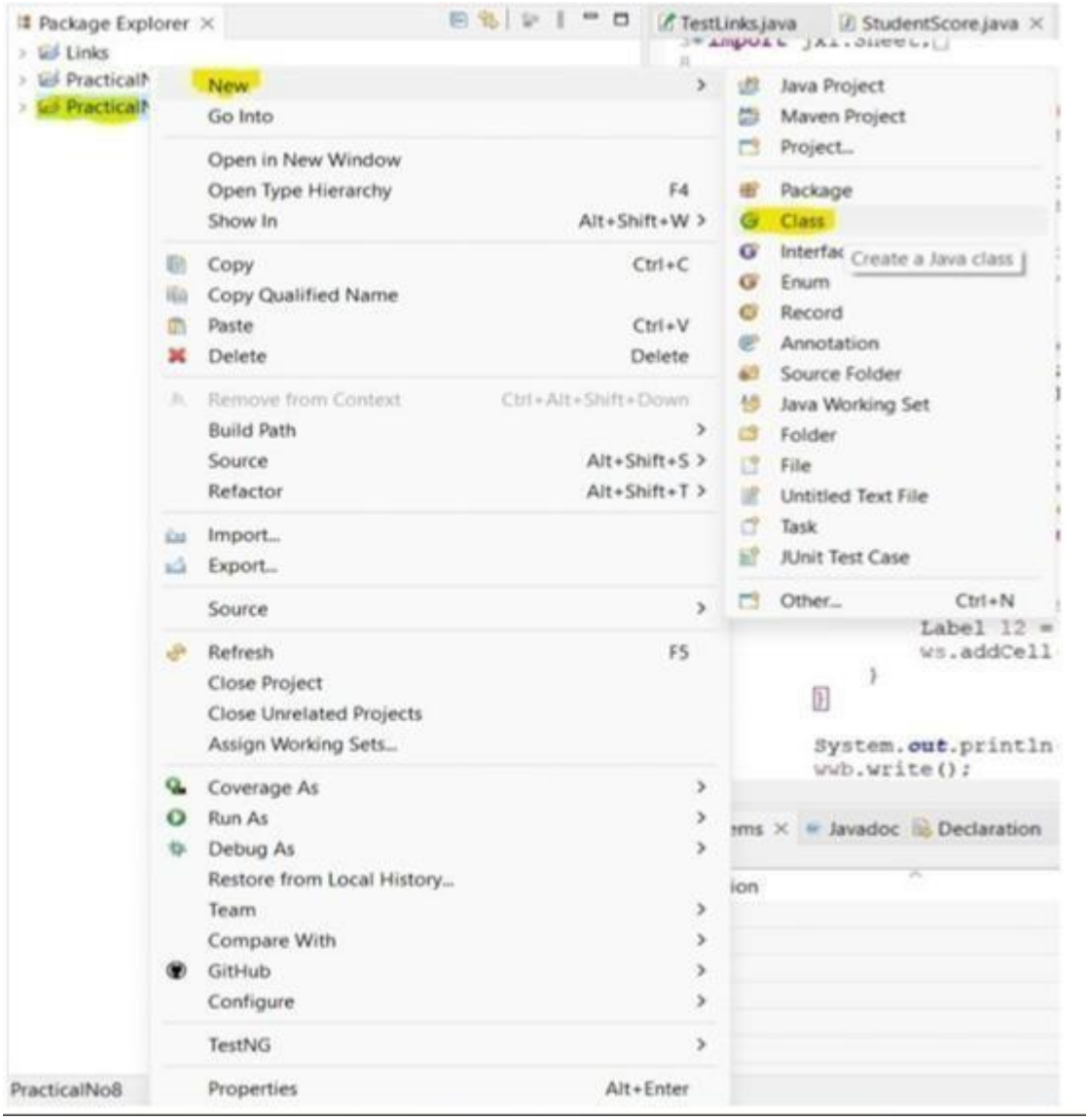
PHP<br>
<input type="checkbox" name="lang" value="C#"> C#<br>
<input type="checkbox" name="lang" value="Python" checked="checked">

Python<br>
<input type="checkbox" name="lang" value="Ruby"> Ruby<br>
<br><br>
<input type="submit" value="Submit">
</form>
</body>
</html>
```

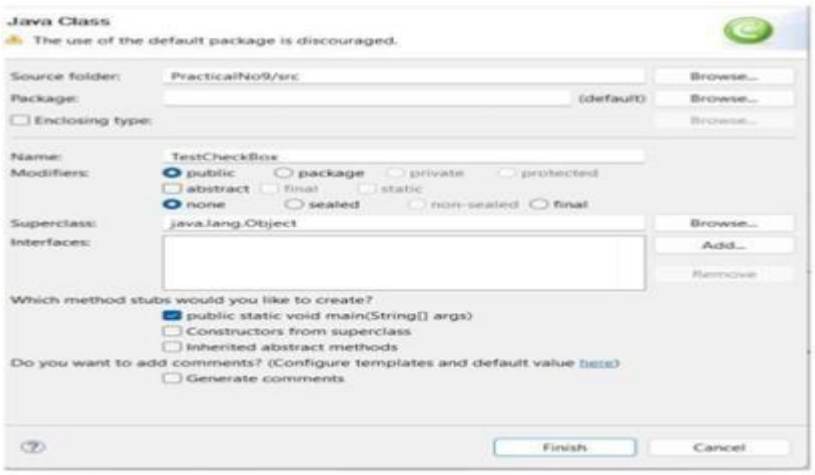
Step 3: Open the Eclipse IDE.
Step 4: Go to File and click New button and Create A Java Project. Give the name on the project.



Step 5: Go to the Left Side Bar and Click the Right button on the project and create a new Class.



Step 6: Then Give the Class Name (TestCheckBox) and Select public static void main(String[] args). Then click on Finish button.



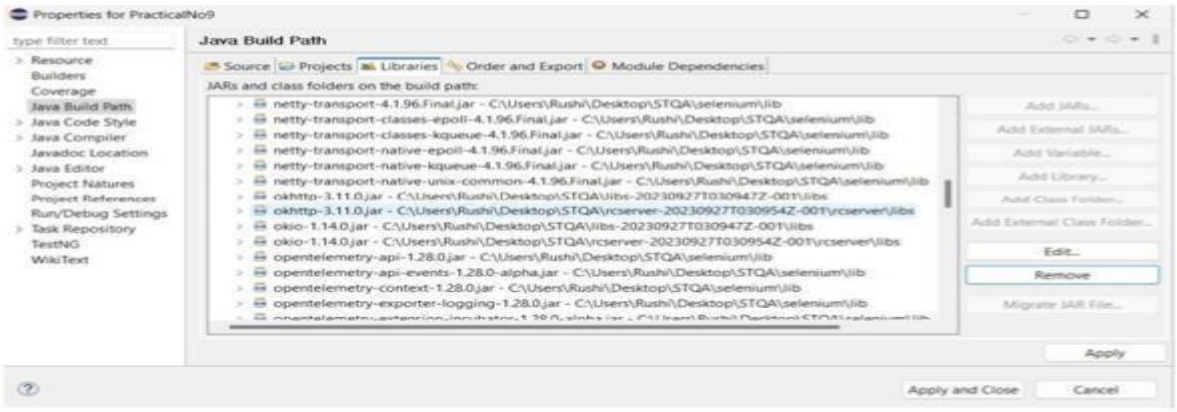
Step 7: Then go again Right Side Click the Right Button On Project Name and click on the Build Path and click on the Configure Build Path.



Step 8: Then Go to Libraries and select the Classpath and Click on the Add External Jars.



Step 9: Add All Jar file. Go to the rcserver, selenium, lib folders and select all the .jar file and Selenium Server Grid file.



Step 10: Copy the HTML code path and the GeckoDriver path to paste in code.

Step 11: Type All Code.

```
import java.util.*; import
org.openqa.selenium.By; import
org.openqa.selenium.WebDriver; import
org.openqa.selenium.WebElement; import
org.openqa.selenium.firefox.FirefoxDriver;


public class TestCheckBox {  public
static void main(String args[])

 {
 System.setProperty("webdriver.gecko.driver","C:\Users\Rushi\Desktop\STQA

\Mozilla Driver-20230927T030950Z-001\Mozilla Driver\geckodriver.exe");
WebDriver driver = new FirefoxDriver();
driver.get("C:\Users\Rushi\Desktop\STQA Practicals\STQA

 PR0.html");  int chk = 0;
int unchk = 0;
List<WebElement> els =

driver.findElements(By.xpath("//input[@type='radio']"));  for(WebElement
el : els)

 {
if(el.isSelected()) {
chk++;
 }

else {
unchk++;  }
 }
```

```
System.out.println("Total checked items" + chk);
System.out.println("Total unchecked items" + unchk);
 }
}
```

Step 12: Run The Code. Step
13: Website page.



Step 14: Write the Code and run it.



```
import java.util.*;

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.firefox.FirefoxDriver;

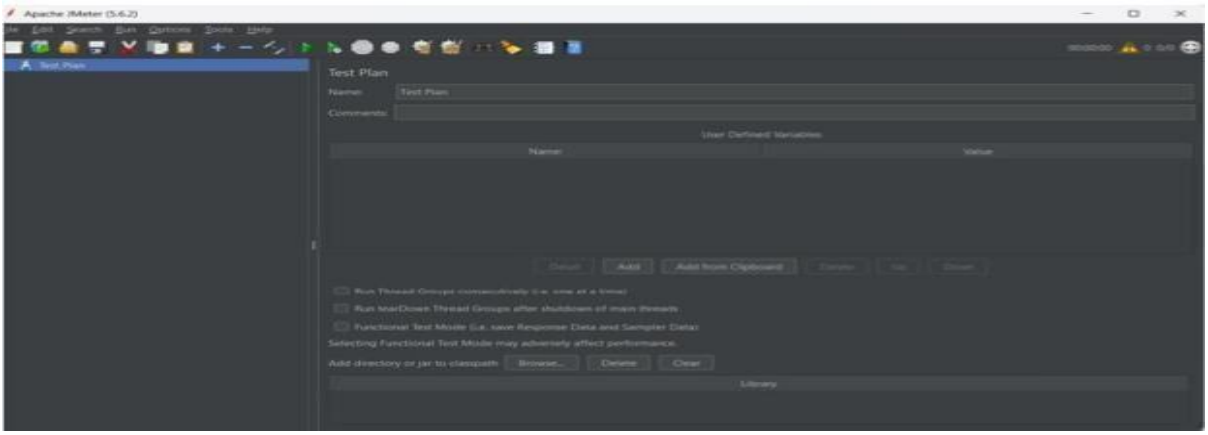
public class TestCheckBox
{
    public static void main(String args[])
    {
        System.setProperty("webdriver.gecko.driver","C:\\Users\\Rushi\\Desktop\\STQA\\Moz
        WebDriver driver = new FirefoxDriver();
        driver.get("C:\\Users\\Rushi\\Desktop\\STQA Practicals\\STQA PRO.html");
        int chk = 0;
        int unchk = 0;
        List <WebElement> els = driver.findElements(By.xpath("//input[@type='radio']"));
        for(WebElement el : els)
        {
            if(el.isSelected()) {
                chk++;
            }
            else{
                unchk++;
            }

        }
        System.out.println("Total checked items" + chk);
        System.out.println("Total unchecked items" + unchk);
    }
}
```

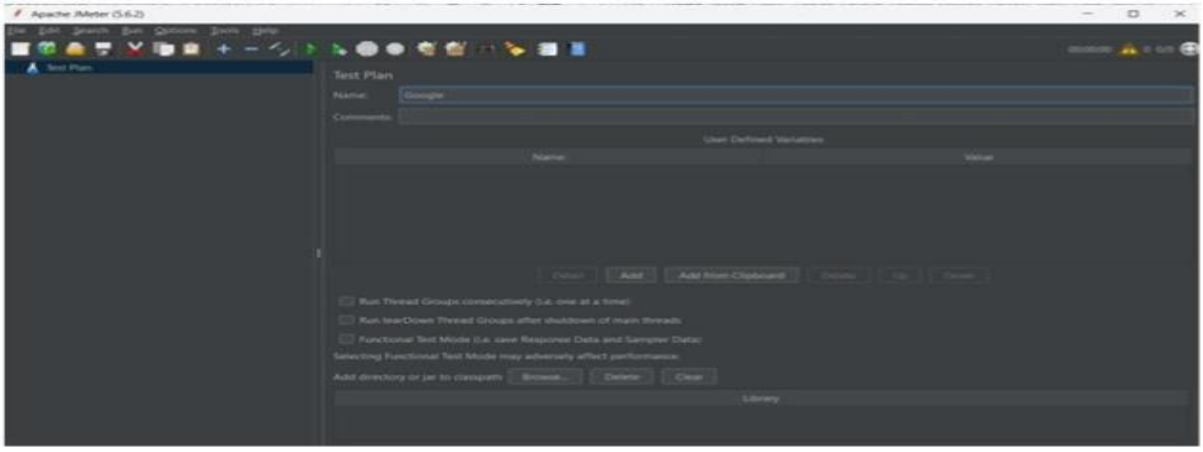
```
Problems  Javadoc  Declaration  Console
terminated> TestCheckBox [Java Application] C:\Program Files\Java\jdk-19\bin\javaw.exe (31-Oct-2023, 1:29:44 pm – 1:29:55 pm) [pid: 262
4983919153?    RemoteAgent    WARN    TLS certificate errors will be ignored for this sessi
avaScript error: resource://gre/modules/XULStore.sys.mjs, line 60: Error: Can't find profile
ct 31, 2023 1:29:54 PM org.openqa.selenium.remote.ProtocolHandshake createSession
NFO: Detected dialect: W3C
otal checked items1
otal unchecked items2
```

PRACTICAL NO. 10
LOAD TESTING USING JMETER.

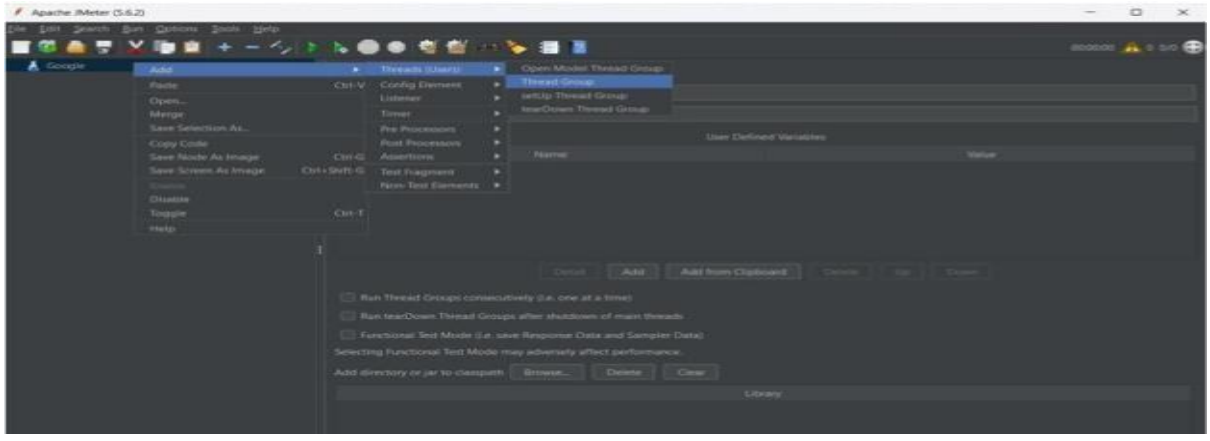
Step 1: Go to The Google And Download Apache JMeter Zip File.
Step 2: Extract The Apache JMeter Zip File.
Step 3: Open the JMeter Folder, then go to Bin Folder and click on Jmeter.sh.



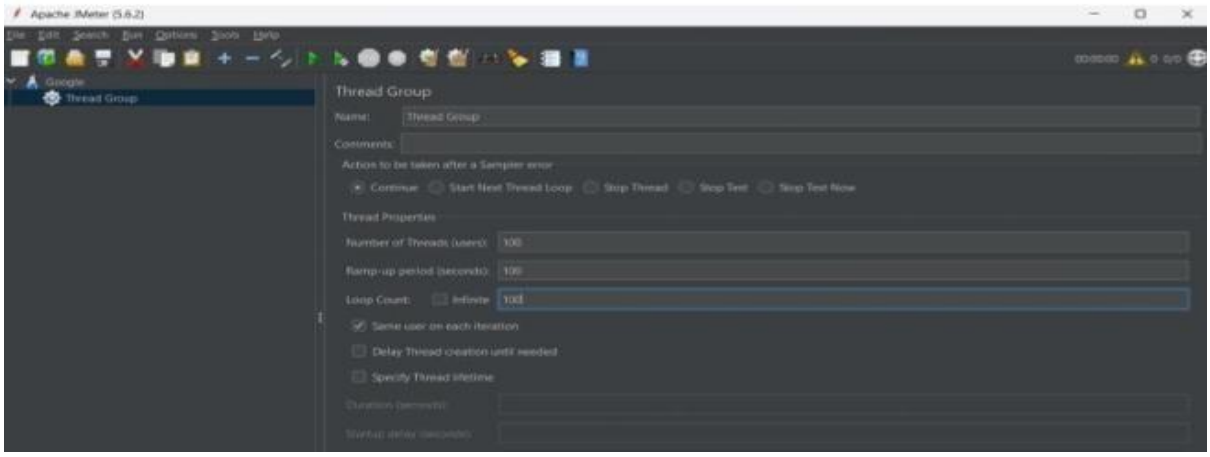
Step 4: Now In JMeter Add A New TestPlan and give the name of TestPlan as “Google”.



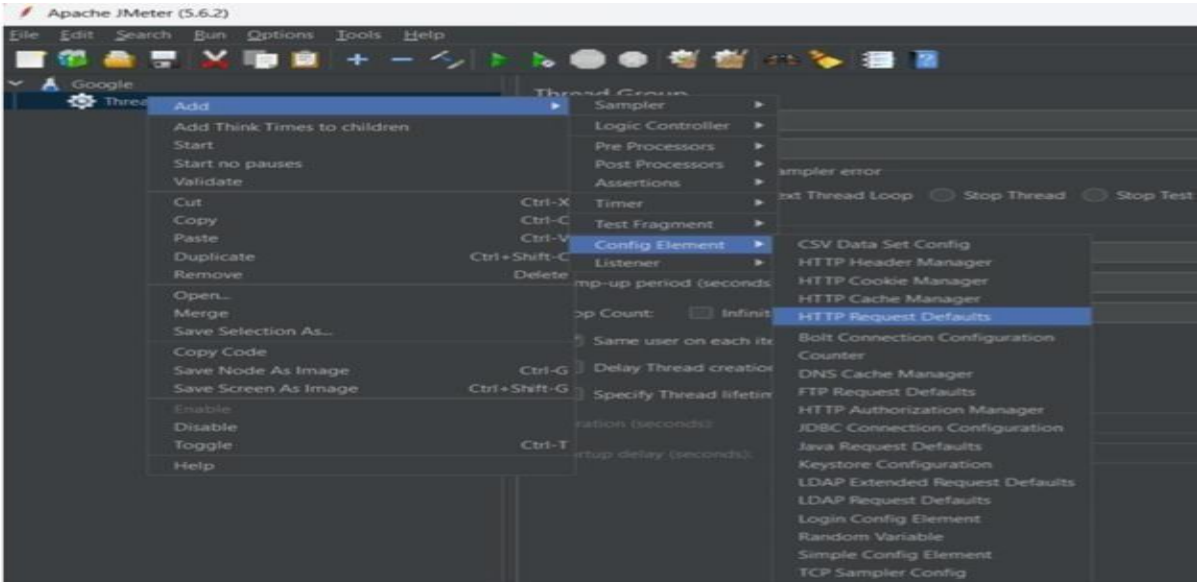
Step 5: Now Right Click On TestPlan. Then Add A Thread and select Thread Group and give the GroupName as “Performance Testing”



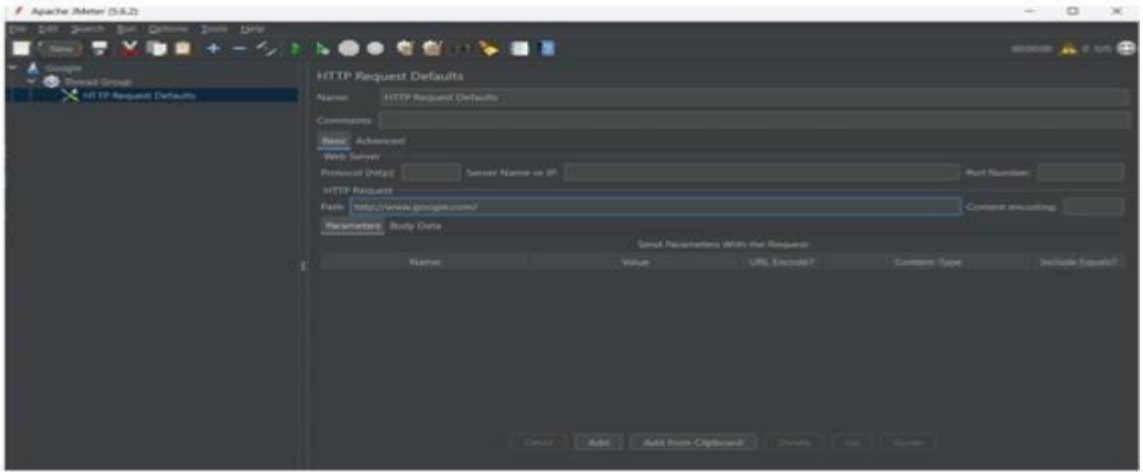
Step 6: Now Add The Number Of Threads, the Ramp-Up Period and Loop control Values as Shown Below.



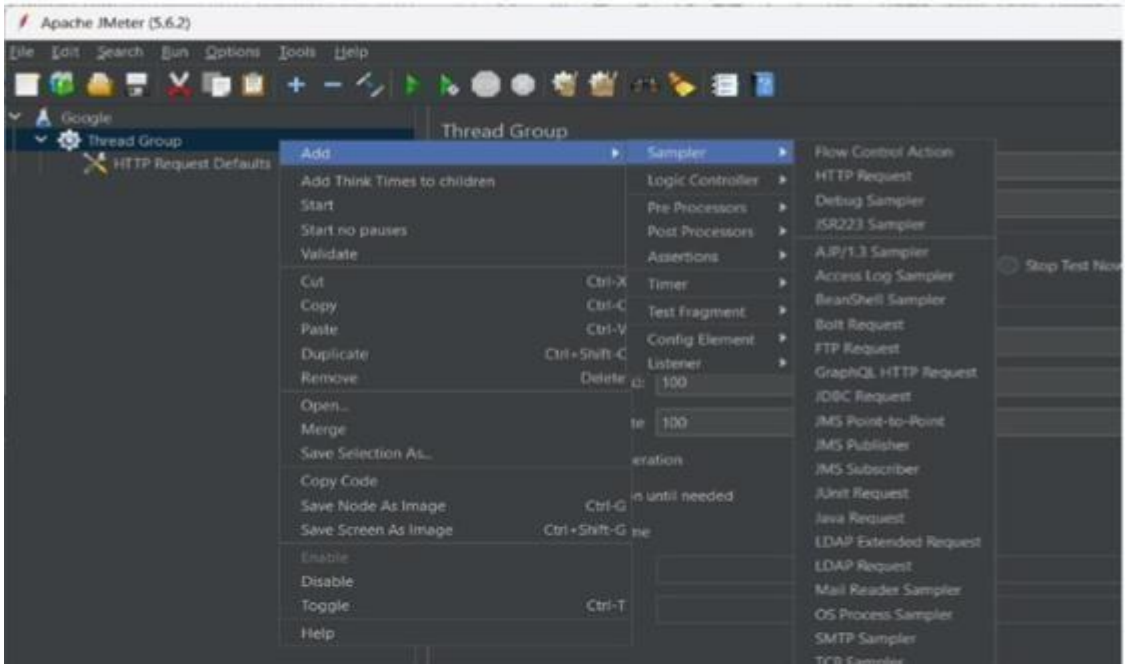
Step 7: Right Click On Thread Group. Add and Select Config Element and Choose The HTTP Request Default.



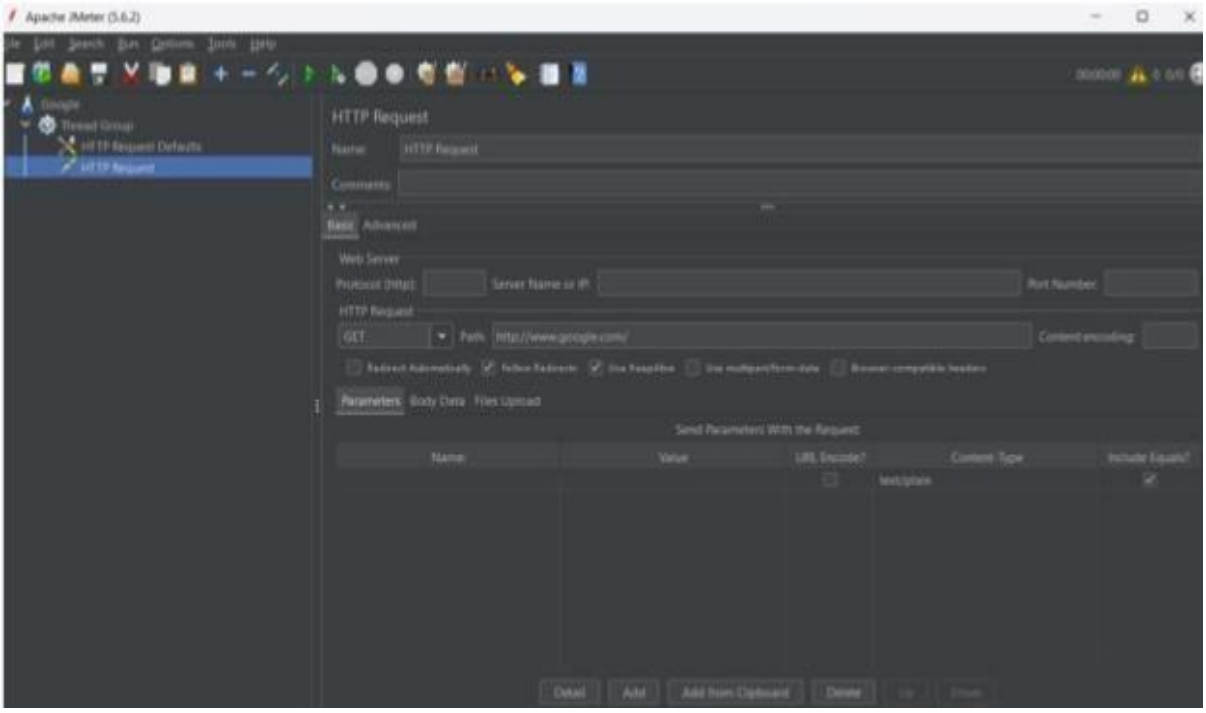
Step 8: Now type http://www.google.com/ in the Path Section.



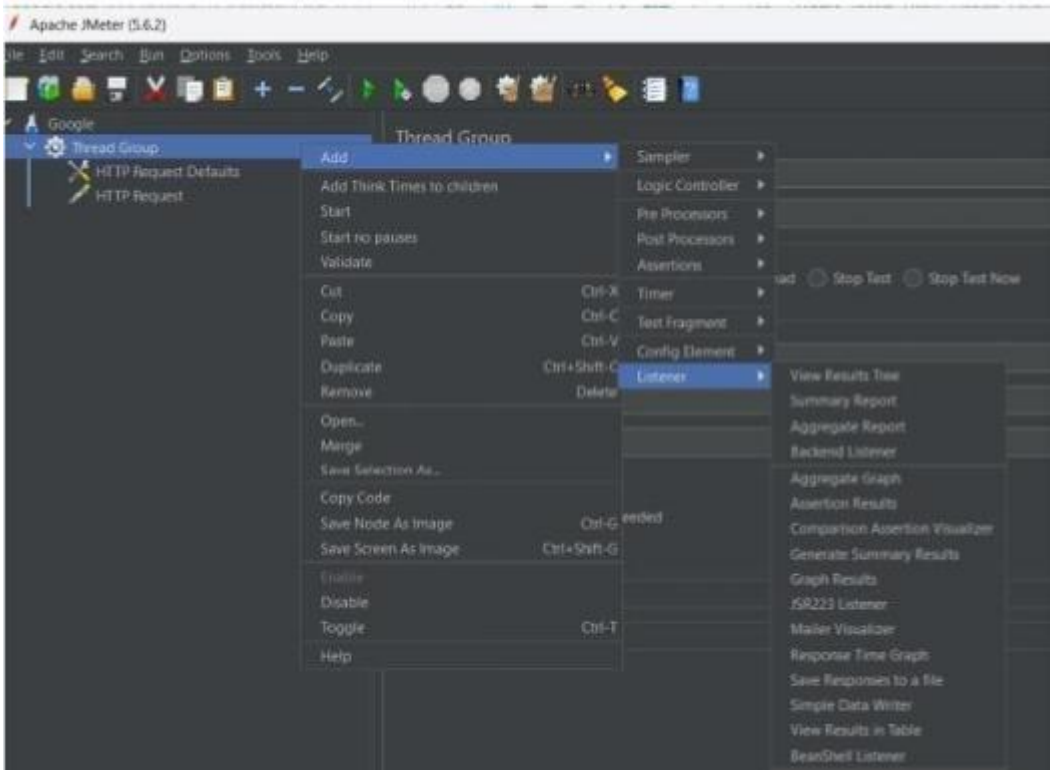
Step 9: Once Again Right Click On the Thread Group, select Add and click on Sampler and choose the HTTP Request.



Step 10: Then Once Again Type The http://www.google.com/ In The Path Section.



Step 11: Once Again Right Click On Thread Group and Add, Listener and Click The Graphics Result.



Step 12: Now Run The Test. And Its Output.

